



Application of artificial intelligence in turbomachinery aerodynamics: progresses and challenges

Zhengping Zou^{1,2} · Pengcheng Xu³ · Yiming Chen^{1,2} · Lichao Yao^{1,2} · Chao Fu^{1,2}

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Abstract

Turbomachinery plays a vital role in energy conversion systems, with aerodynamic issues being integral to its entire lifecycle, spanning the period of design, validation, and maintenance. Conventionally, the reliance on skilled aerodynamic engineers has been pivotal in the successful development of turbomachines. However, in the current era of burgeoning artificial intelligence (AI) technology, researchers are increasingly turning to AI to replace human expertise and decision-making in these aerodynamic issues and to solve previously intractable aerodynamic problems. This paper presents a systematic literature review of the latest advancements in applying AI to turbomachinery aerodynamics, encompassing the design, validation, and maintenance of compressors and turbines. It underscores how AI is revolutionizing the research paradigm of turbomachinery aerodynamics. AI's powerful learning capability facilitates more precise and convenient aerodynamic analyses and inspires innovative aerodynamic design ideas that go beyond the capabilities of classical design techniques. Additionally, AI's autonomous decision-making capability can be employed for aerodynamic optimization and active flow control of turbomachines, generating optimal aerodynamic solutions and complex control strategies that surpass human brains. As a main contribution, we provide a detailed exposition of the future intelligent turbomachinery research and development (R &D) system, along with highlighting potential challenges such as physics embedding, interactive 3D design optimization, and real-time prognoses. It is anticipated that harnessing AI's full potential will lead to a comprehensive AI-based turbomachinery R &D system in the future.

Keywords Artificial intelligence · Turbomachinery · Machine learning · Aerodynamic design · CFD · Maintenance

✉ Zhengping Zou
zouzhenping@buaa.edu.cn
Pengcheng Xu
xupengcheng@buaa.edu.cn

¹ National Key Laboratory of Science and Technology on Aero-engine Aero-thermodynamics, Beihang University, Beijing 100191, China

² Research Institute of Aero-engine, Beihang University, Beijing 102206, China

³ Taihang Laboratory, Chengdu 610000, China

1 Introduction

The increasing global demand for renewable, sustainable, and environmentally friendly electricity is being addressed through a broad and fragmented array of green energy technologies. Solar, wind, hydro, tidal, wave, geothermal, and biomass energy resources necessitate advanced turbomachinery designs to efficiently extract power from low-density energy. Virtually almost all refined, new, and proposed technologies for generating green electricity rely on improved aerodynamic designs for turbines, compressors, expanders, pumps, and fans to make them economically and environmentally viable (Wang et al. 2020). Therefore, turbomachinery plays a vital role in sustainable development.

Turbomachines are devices in which the aerodynamic action of steadily rotating blade rows transfers the work to or from a fluid. The aerodynamic action is important, as it makes the energy transfer, pressure, and momentum change a reality in turbomachinery. The aerodynamics involved in turbomachinery belong to the subcategory known as internal flows. In the turbomachinery internal aerodynamics, the fluid alternately flows through stationary and rotating confined passages which are defined by the shape of blades and endwalls, and finally these intricate unsteady three-dimensional (3D) flows facilitate energy conversion within the turbomachines.

The research and development (R &D) of turbomachinery revolves around the turbomachinery aerodynamics, with a central focus on the study of the relationship between “geometry, internal flow, and aerodynamic performance”, as shown in Fig. 1. The research can be categorized into two phases:

In the aerodynamic design phase, aerodynamic researchers primarily concentrate on establishing connections between deterministic geometry and aerodynamic performance. Modern aerodynamic design of turbomachinery is an incremental design-optimization process from low to high dimensions. In each dimension, the design process entails selections of velocity triangles, blade profiles, or stacking curves based on the designer’s prior experience, whereas the optimization process involves performance predictions with the aid of cutting-edge analysis tools and geometrical refinements by skilled designers. A comprehensive design system and a sophisticated designer are the two foundations of aerodynamic design.

In the operation and maintenance phase, aerodynamic researchers primarily focus on establishing the connection between manufactured geometry and aerodynamic performance in real operating conditions. During the operation, the aerodynamic state of turbomachines inevitably undergoes changes due to various factors such as manufacturing deviations, inflow fluctuations, and blade defects. Condition monitor is necessary because prognoses and diagnoses can be performed based on the monitoring data of aerodynamic state, and maintenance should be conducted based on the operability prediction. Therefore, experienced aerodynamic researchers must be deeply involved in this phase.

In summary, from an aerodynamic perspective, the successful development of a turbomachine necessitates both an advanced R &D system and an experienced aerodynamic researcher. These requirements significantly elevate the threshold for turbomachinery development. On one hand, the evolution of such an R &D system relies heavily on substantial practice and distilled insights, as well as continuous investment to ensure timely upgrades. On the other hand, nurturing an experienced designer demands a substantial investment of time and resources. Moreover, the transfer of personal experience during the mentoring process might lead to certain aspects being overlooked, causing novice designers to expend considerable effort on trivial tasks that were initially unnecessary.

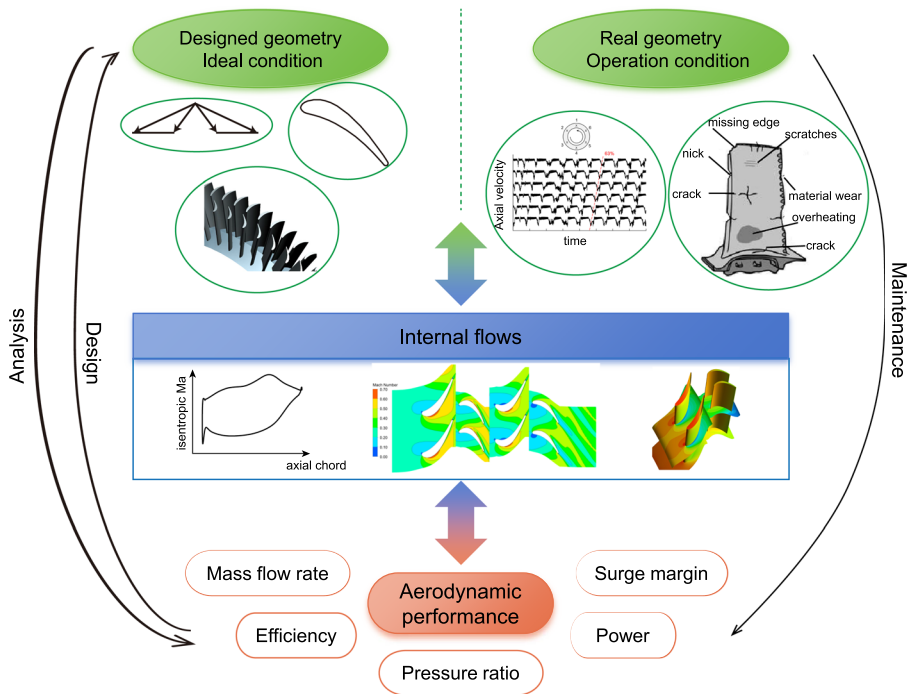


Fig. 1 The core for turbomachinery aerodynamics is the relationship between “geometry, internal flow, and aerodynamic performance”

The essence of the above two thresholds lies in the realm of knowledge acquisition and transmission. Historically, addressing such challenges has solely relied on human intelligence. In recent years, the rapid development of AI has introduced an entirely new avenue for tackling these challenges. AI is a novel technological discipline focused on the development of theories, methods, techniques, and applications for simulating and extending human intelligence (Russell and Norvig 2009). AI possesses the capacity for knowledge collection, extraction, expansion, and expression. It not only allows for the assimilation and summarization of past design experiences into its knowledge repository, thus reducing reliance on domain-specific expertise, but also actively mines new knowledge based on fresh data. This enables autonomous expansion and updates of the knowledge system, aligning it with the ever-evolving technological landscape.

In this context, the integration of AI with research in turbomachinery aerodynamics heralds some new possibilities. Over the past few years, more and more researchers have been engaged in exploring the utilization of AI in the entire lifecycle of turbomachinery, from design and validation to maintenance. As evidenced by the growing number of publications, a comprehensive review of the relevant literature is timely and necessary.

This paper provides a systematic review for the application of AI in turbomachinery aerodynamics. We begin by reviewing the evolution of the research paradigm in the field of turbomachinery aerodynamics and, in conjunction with the characteristics of aerodynamic tasks, introduce the potential transformative effects driven by AI technology. The ensuing three sections comprehensively delve into recent representative works in AI applied to

turbomachinery aerodynamics at the phases of aerodynamic design, validation, and maintenance. Based on these efforts, this paper summarizes the main breakthrough directions facilitated by AI in turbomachinery aerodynamics. Finally, we try to elucidate the holistic profile of a next-generation AI-based turbomachinery R & D system, characterized by features such as knowledge acquisition, active learning, and self-expansion. Intending to accelerate the establishment and development of such a system, this paper also identifies current research limitations and offers insights into potential future directions.

2 Characteristics of turbomachinery aerodynamics from a data perspective

This section outlines the evolutionary trajectory of research paradigms in turbomachinery aerodynamics, summarizes the characteristics of aerodynamic tasks in the age of data, and provides a brief overview of some potential machine-learning methods that could be harnessed for these tasks.

2.1 Development of research paradigm for turbomachinery aerodynamics

The research of turbomachinery aerodynamics has undergone three distinct phases: “experience induction—model derivation—numerical simulation”. Taking the primary aerodynamic task, aerodynamic design, as an example, Fig. 2 summarizes the key techniques that occur in each phase. Almost half a century ago, turbomachinery designs were primarily based on empirical correlations such as Smith chart (Smith 1965), Cordier diagram (Cordier 1955), and Balje diagrams (Balje 1981). Blade profiles could only be selected from prescribed airfoils like NACA 65 Series. By the late 1970s, with the maturity of quasi-three-dimensional (Q3D) design systems developed from Wu’s theory (Wu 1952), designers started employing more advanced profile-shaping techniques by controlling the circulation (Zangeneh 1991; Dang and Isgro 1995; Demeulenaere et al. 1997). The emergence of the controlled diffusion airfoil (Hobbs and Weingold 1984) was a hallmark achievement of this era. In the early 1980s, computational fluid dynamics (CFD) gradually became a significant design tool that enabled full 3D detailed design (Denton and Xu 1998; Horlock and Denton 2005). A series of advanced 3D design technologies emerged one after another, such as swept blades (Wennerstrom 2001; Rosic and Xu 2012), non-axisymmetric endwalls (Atkins 1987), squealer tips (Zou et al. 2017, 2020), etc. With the continual enhancement of computing power, designers further developed unsteady design techniques that considered unsteady phenomena like the Calming effects (Curtis et al. 1997; Liang et al. 2015) in multi-stage turbomachines, as well as robust design techniques that considered geometric (Wang and Zou 2019; Wang et al. 2022c) and operation uncertainty (Luo et al. 2021). The aforementioned three phases correspond to three paradigms in scientific study: “empirical science—theoretical science—computational science” (Hey 2009). The modern turbomachinery aerodynamic design system is an integrated application of the above three paradigms (Zou et al. 2018), as demonstrated by the three design & optimization steps in Fig. 3.

The evolution of research paradigms continually gives rise to new techniques. The fourth paradigm, data-intensive science (Hey 2009), is coming with the advent of AI era. This shift transforms the focus from computation-centric to data-centric, and data-driven

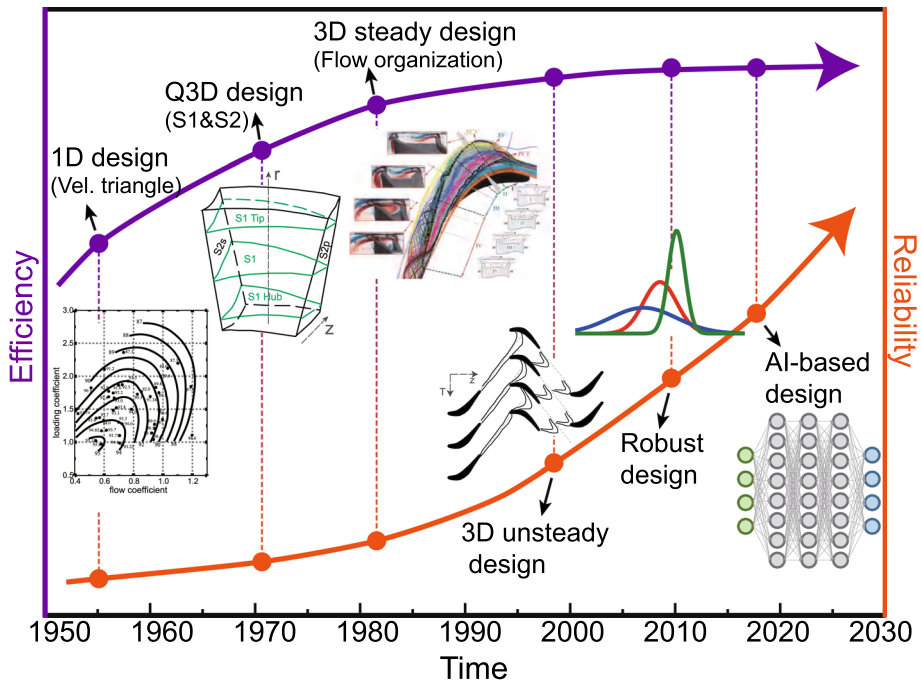


Fig. 2 Paradigm development of the aerodynamic design for turbomachinery. There are four distinct phases in the research of turbomachinery aerodynamics: “experience induction — model derivation — numerical simulation — data science”. Current developments in related technologies are still in the early stages of the fourth phase

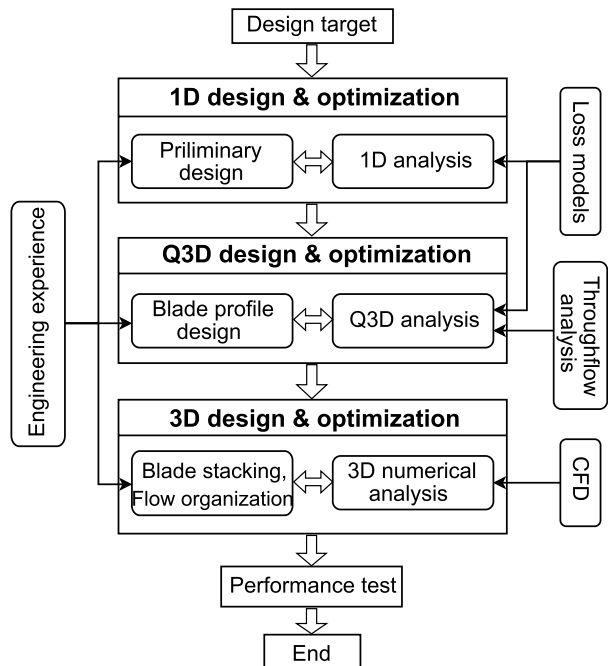
methods hold the potential to become the primary means of future aerodynamic research on turbomachinery.

2.2 Characteristics of aerodynamic tasks in turbomachinery

From the perspective of data, the characteristics of aerodynamic tasks in turbomachinery can be summarized as the following three aspects:

1. **High dimensions.** Turbomachinery inherently possesses a complex three-dimensional geometry, with internal flows containing numerous intricate elements including leakages, separations, and transitions. An accurate description of all the flow information necessitates a high-dimensional spatial representation.
2. **Big data.** In the past century, turbomachinery has been widely applied in aerospace, vehicles, ships, energy conversion, and other industrial fields. Plentiful data has been accumulated in the process of design, tests, and operations. Meanwhile, fresh data is increasingly being produced.
3. **Compliance with scientific laws.** Aerodynamic tasks in turbomachinery are essentially scientific issues that rigorously adhere to all physical laws of aerodynamics and thermodynamics.

Fig. 3 Modern aerodynamic design system for turbomachinery. The most advanced turbomachinery aerodynamic design system at present is an amalgamation of the technologies from the first three phases. With the maturation of technologies in the fourth phase, it holds the promise of a revolutionary upgrade to the current aerodynamic design system



The aforementioned attributes of high dimensions and big data make aerodynamic tasks in turbomachinery well-suited for AI's applications, while the requirement of rigorous compliance with physical laws introduces certain intricacies that require particular attention during the implementation.

2.3 Potential machine learning methods for turbomachinery aerodynamics

Machine learning is the leading pathway to AI. AI uses machine learning to accomplish most aerodynamic tasks in turbomachinery. Table 1 summarizes some aerodynamic tasks

Table 1 Application of machine learning in typical aerodynamic tasks of turbomachinery

	Tasks	Functional requirement	Algorithms
Aerodynamic design	Feature extraction (Pongetti et al. 2021)	Dimension reduction	PCA, AE
	Loss modelling (Yue et al. 2022)	Regression	KNN
	Loss identification (Angelini et al. 2021)	Clustering	K-means
	Performance prediction (Thatte et al. 2021)	Regression	FNN
	Flow field prediction (Li et al. 2022b)	Regression	Dual GNN
	Optimization (Liu et al. 2022)	Decision making	RL
Maintenance	Stall warning (Lin et al. 2019)	Classification	FNN
	Defect detection (Talebi et al. 2022)	Anomaly detection	FNN
	Operability prediction (Raghavan et al. 2020)	Regression	FNN

that have been addressed using machine learning in published literature, along with their functional requirements and corresponding methods.

Regression and classification. Regression and classification are the most common problems in turbomachinery aerodynamics. They are both predictive problems that differ only in the type of predicted data, and thus both can be solved by supervised learning algorithms. Numerous algorithms have been developed for them, such as K-Nearest Neighbor (KNN), Gaussian process (GP), Support vector machine (SVM), Random Forest, and Neural networks (NN). A summary of their basic principles, as well as advantages and disadvantages, is provided in Table 2. NN deserves special attention because it is really popular today and has been developed into a large family, including Fully-connected neural networks (FNN), Convolutional neural networks (CNN), Graph neural networks (GNN), Conditional generative adversarial networks (cGAN), etc. In practical applications, it is necessary to choose the appropriate machine learning method based on the characteristics of specific aerodynamic tasks.

Dimensionality reduction. Dimensionality reduction refers to the transformation of data from high-dimensional space to low-dimensional space while preserving the maximum variance in the data. It can alleviate the sparsity issue of data samples in high-dimensional space, provide initial denoising of data, and also offer a more thorough or insightful way to describe the data, thus facilitating subsequent data analysis. Commonly used algorithms include principal component analysis (PCA) (Hotelling 1933) and auto-encoder (AE). PCA converts a set of linearly correlated variables into a new set of linearly independent variables using orthogonal transformation, thus presenting the data in a smaller dimension. It is simple and convenient but the number of principal components generated is uncontrollable. AE trains two sets of neural networks (Encoder and Decoder) supervised by the input data itself. A set of latent parameters can be obtained by the encoder, while the decoder can use them to reconstruct the original data. AE is more flexible than PCA, with a controlled number of latent parameters and the ability to handle nonlinearly correlated features, but it is less interpretable. In the field of turbomachinery aerodynamics, dimensionality reduction methods are commonly applied to preprocess high-dimensional data such as 3D blade geometry and 3D flow fields. This helps reduce the complexity of subsequent operations, such as relationship mapping and flow field reconstruction.

Clustering. Clustering is an unsupervised machine learning method that divides the data into clusters according to shared characteristics. This method can automatically distinguish data that is challenging to manually characterize, and each cluster potentially corresponds to an underlying concept. Clustering is ideally suited for identifying special flows in turbomachines that are highly characteristic but difficult to define mathematically, such as boundary layers, shock waves, secondary flows, etc. Researchers (Angelini et al. 2021) have used the classical clustering algorithm K-means (Hartigan and Wong 1979) to identify the boundary layers and wake regions in axial compressors.

Decision making. AI is able to make decisions through reinforcement learning (RL). During the RL, an agent makes actions based on its state and receives an immediate or delayed reward from the environment (Sutton and Barto 1998). The agent learns the policy from its interactions with the environment on target to maximize the desired total reward. In more complicated problems, DNN can be used to describe the policy function or value function, which leads to a series of deep reinforcement learning algorithms such as Deep Q learning (Mnih et al. 2015), Trust region policy optimization (Schulman et al. 2015), and Deep deterministic policy gradient (Lillicrap et al. 2019). In turbomachinery, RL is appropriate

Table 2 Brief introduction of frequently-used machine learning algorithms in turbomachinery aerodynamics

Method	Principle	Pros	Cons
KNN (Cover and Hart 1967)	It assumes that similar data points can be found near each other and prediction can be performed based on its proximity and association to other available data	Simple and convenient to implement. No need for repeating training if new data are added. Non-parametric. Only one hyperparameter k is required	Computationally expensive for large datasets. Cannot handle data with missing values. Requires large memory to store all data. Tends to fall victim to the curse of dimensionality and overfitting
GP (Rasmussen and Williams 2005)	GP (sometimes called kriging) is a collection of random variables such that the joint distribution of every finite subset of random variables is multivariate Gaussian	Convenient to implement. Non-parametric. Flexibility to add prior knowledge and specifications about the shape of the model by selecting different kernel functions Estimation of uncertainty in a principled way	Selection of kernel functions depends on engineering experience Less efficient for large datasets
SVM (Cortes and Vapnik 1995)	It maps the data to higher dimensions by kernel functions and then constructs a hyperplane where the distance between two classes of data points is at its maximum	Works well for both high- and low-dimensional data Possesses solid mathematics theoretical basis. Can well handle nonlinear mapping and has excellent generalization performance Fast prediction	Less effective with large datasets The accuracy depends on the selection of kernel functions and tuning of hyperparameters Needs careful pre-processing including feature extraction and data normalization Difficult to explain the results
Random Forest (Breiman 2001)	It is a collection of uncorrelated decision trees which are merged together to reduce variance and create more accurate data predictions	Easy data preparation without any need for normalization or scaling Robust to noise, outliers, and missing values Easy for parallel processing Lower risk of overfitting Partly explicable, and easy to determine feature importance	Time-consuming for large datasets Limited with regression especially when data has linear nature Less intuitive if too many decision trees are employed Biased towards categorical variables with more levels

Table 2 (continued)

Method	Principle	Pros	Cons
DNN (Rosenblatt 1958; LeCun et al. 1989)	Primarily leveraged for deep learning (Deep neural networks, DNN). It is a parallel information processing system consisting of multiple layers of artificial neurons. Neurons are fine-tuned by back-propagating errors to fit the complicated nonlinear mapping relation	State-of-the-art performance Strong representation ability, a sufficiently large neural network can approximate any continuously differentiable functions No need for feature engineering Easy for parallel processing Fast prediction. Good transferability for transfer learning	Black box with low interpretability Computationally expensive Requires careful design for the network structure Requires an adequate amount of labeled data

for issues of optimization and active control of turbomachines (Jiang et al. 2020; Parvaresh et al. 2020; Liu et al. 2022).

Advanced learning paradigm. Some advanced learning paradigms, such as transfer learning and lifelong learning, may also be used in the application of AI. Transfer learning applies knowledge learned from the source domain to a related problem in the target domain, making it possible for reliable learning on small samples (Pan and Yang 2010; Pan et al. 2011). However, knowledge transfer is unidirectional. Fine-tuning the pre-trained model destroys the learned weights, consequently leading to poor inference when the model is in turn applied to the source domain. This problem is also known as catastrophic forgetting (Kirkpatrick et al. 2017). To address the problem, researchers (Thrun 1998) proposed lifelong learning. Lifelong learning leverages past knowledge to help learn new tasks, while all the knowledge is accumulated and stored in a knowledge base. In other words, lifelong learning has the ability to learn new knowledge without catastrophic forgetting.

3 AI-based aerodynamic design system

A framework for the AI-based aerodynamic design system is illustrated in Fig. 4. This system represents a groundbreaking upgrade to the modern aerodynamic design system in the Fig. 3b. It leverages AI's knowledge extraction capabilities to replace the need for traditional empirical expertise and numerical methods in the design analysis process. It also employs AI's autonomous decision-making abilities to supplant conventional optimization algorithms in the optimization process. The AI-based aerodynamic design of turbomachinery is a process of incremental design and optimization from low to high dimensions. In each dimension, AI creates an initial geometry via knowledge learned from the database, and then uses it as a starting point for optimization to improve aerodynamic performance. Therefore, this section will review the application of AI from the perspectives of design and optimization.

3.1 Design mode

During the design process, AI generates high-dimensional geometric data from low-dimensional design targets, based on design models learned from databases of well-designed turbomachines. Its functionality is similar to traditional inverse design methods in the field of turbomachinery. Traditional inverse design methods include permeable-boundary methods (Zangeneh 1994; Dang and Isgro 1995; Demeulenaere et al. 1997) and virtual-wall-movement methods (Thompkins and Tong 1982; Daneshkhah and Ghaly 2007). Both methods involve special mathematical treatments of the blade surface boundary conditions in Euler or Navier–Stokes solvers to determine the relationship between pressure distribution and blade shape. These inverse problem methods, although effective in reducing the design cycle by eliminating unguided iterative blade modifications, suffer from decreased solving stability due to the special treatment of governing equations. Additionally, inappropriately prescribed loading distribution can lead to ill-posed problems where a physically corresponding blade geometry does not exist. By contrast, AI-based inverse design models are trained by databases obtained from more stable forward analyses, thus avoiding convergence issues. Moreover, AI can provide optimal loading distributions for inverse design, which helps avoid ill-posed problems caused by inappropriate input. This capability is not possessed by traditional inverse design methods. This section will specifically introduce the work in this regard.

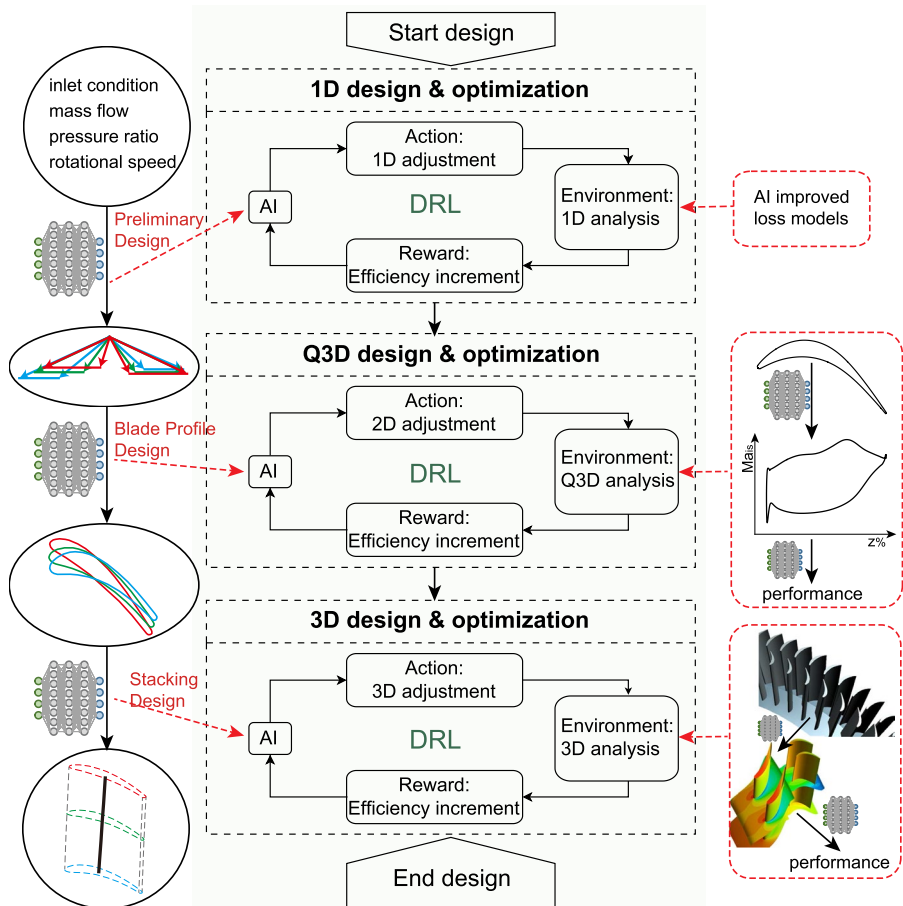


Fig. 4 Framework for the AI-based aerodynamic design system. It is an upgrade of the modern aerodynamic design system in Fig. 3b. It leverages AI's knowledge extraction capabilities to replace the need for traditional empirical expertise and numerical methods in the design-analysis process. It also employs AI's autonomous decision-making abilities to supplant conventional optimization algorithms in the optimization process

Preliminary design

The purpose of the preliminary design is to determine the velocity triangles and the contour of the meridional flow path (Xu et al. 2022). While in the past these tasks were typically solved by aerodynamic designers based on available design guidelines and engineering experience, now AI can inherit this knowledge through machine learning. AI extracts design knowledge from a state-of-the-art database and then applies it to new design projects. This application can be categorized into two approaches: direct design and indirect design.

Direct design refers to direct application of the mapping between design targets to velocity triangles from historical databases in the new design task. Models established under this approach inherit excellent design features from past instances, enabling the new turbomachine to start from a higher point. But unfortunately, these features are recorded

in the models in a black-box manner. A feasible 1D aerodynamic design solution has been proposed by the author's team (Chen and Liu 2022).

Indirect design refers to the process of extracting design guidelines from a database and then manually applying these guidelines to the preliminary design. The idea is similar to the generation of the traditional Smith chart (Smith 1965) and $n_s - D_s$ diagram (Balje 1981). However, unlike traditional methods that simplify the design freedom to only two dimensions, the exceptional capability of AI in handling high-dimensional information allows the generated design guidelines to contain more information. Taking an axial fan as an example, Angelini et al. (2019) analyzed 7313 sample points on the $n_s - D_s$ diagram using principal component analysis (PCA) and projection to latent structure (PLS). The first principal component is added to the traditional $n_s - D_s$ diagram to guide the selection of hub-to-tip ratio and solidity. This increases the dimensionality of the $n_s - D_s$ diagram, enriches its content, and expands its application scope.

Among the two AI application approaches mentioned above, direct design is a method that relies entirely on AI's knowledge transfer capability. It exhibits strong autonomous decision-making ability, and its applicability expands as the database grows. However, due to its black-box nature, this method has limited interpretability. On the other hand, indirect design uses AI to improve and supplement traditional methods. Such methods generate design guidelines that assist designers in gaining a better understanding of the rationale behind a particular design. However, during the design process, more manual decision-making is required.

Blade profile design

In this step, AI defines a smooth profile that satisfies the basic restrictions imposed by the result of preliminary design. Currently, research efforts are primarily concentrated in the field of airfoil design for aircraft, with very limited research focused on turbomachinery. We (Chen et al. 2022) have proposed a two-stage model for this purpose based on deep learning and multi-output Gaussian process (MOGP). Figure 5a illustrates its application in an axial turbine. In the first stage, a DNN-enhanced MOGP model is built to identify the optimal loading distribution based on the design targets. In the second stage, a DNN model is developed to generate the turbine profile according to the optimal loading distribution. This formulation enables the automatic determination of the optimal turbine geometry without the need for an iterative process. Moreover, the knowledge within the design system can expand as the database grows.

The authors have used this framework to redesign a low-pressure turbine cascade (Pak-B) from Pratt and Whitney (Praisner et al. 2013). The optimal loading distribution is predicted according to the design condition of Pak-B, and the corresponding blade profile (redesigned) is generated as shown in Fig. 6a. Performance of these two profiles are compared through both CFD and experiments. It is found that the aerodynamic performance of the redesigned profile is considerably better than that of Pak-B under all Reynolds numbers, which demonstrates the effectiveness of this AI-based blade profile design framework.

3D blade design

After determining optimal profiles with minimal loss at the blade root, mid, and tip, the next step involves identifying a proper stacking curve to construct the optimal 3D blade. The authors have also provided an AI-based solution for 3D aerodynamic design. It is actually a DNN mapping from velocity triangles to corresponding optimal stacking curves which can minimize the secondary loss. Taking the axial turbine as an example,

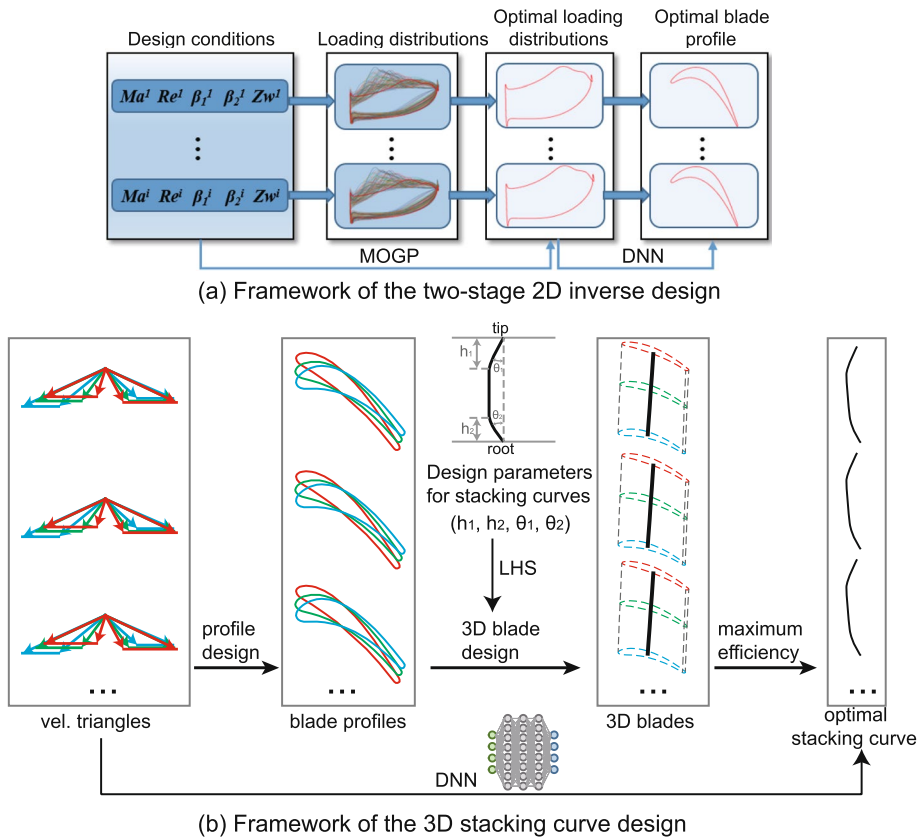


Fig. 5 Framework of the 2D and 3D inverse design (Chen et al. 2022). **a** The two-stage model for 2D inverse design of axial turbines. **b** Procedures to find the optimal stacking curve for a given set of velocity triangles

Fig. 5b demonstrates the database creation process. Firstly, sufficient representative velocity triangles are randomly selected. Secondly, for each set of velocity triangles, blade profiles are designed by AI models developed in the previous subsection. Thirdly, alternative stacking curves are created by randomly sampling design parameters, and then various 3D blades are consequently constructed. Finally, numerical simulations are performed for all the 3D blades, and the stacking curve with maximum efficiency is selected as the optimum. This method has been employed to redesign the first stage of the E3 low-pressure turbine from General Electric (Cherry 1982). After inputting the original velocity triangles, the 3D blade can be built in 7 s by AI. As can be seen in Fig. 6b, AI can efficiently generate 3D blades with equivalent performance to that produced by experienced designers.

Ghosh et al. (2021) offers an alternative solution in the form of 3D inverse design. It is called a probabilistic inverse design machine learning framework. It uses a conditional invertible neural network to design the desired 3D blade explicitly from design targets of performance and constraints. The input parameters are the target efficiency and spanwise distribution of pressure, flow angle, and degree of reaction. The output is the 3D blade geometry. There is no need for separate blade profile design, but appropriate pressure and

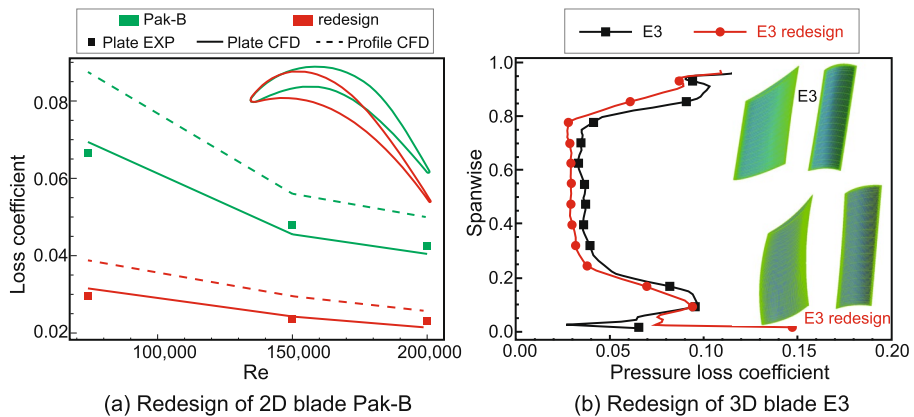


Fig. 6 Redesign results of a 2D turbine cascade and a 3D blade (Chen et al. 2022). **a** Comparison of the loss coefficient of Pak-B and the redesigned profile. ‘Plate’ means results measured from the boundary layer on the flat plate with the same loading distribution of the profile. **b** Comparison of spanwise pressure loss coefficients of the prototype and redesign E3 stage

swirl angle profiles should be input as prior knowledge. How to obtain the best pressure profile is still a challenge.

3.2 Optimization mode

During the optimization process, AI first evaluates the designed geometry and then tries to make some refinements. Evaluations and refinements are iteratively employed in an effort to achieve the desired aerodynamic performance. It is essential that these evaluations strictly adhere to the principles of turbomachinery aerodynamics to ensure the reliability of the results.

1D analysis

The most straightforward application of AI in 1D performance analysis is to establish a mapping from 1D geometric parameters to the aerodynamic performance, similar to the work of Ren et al. (2022) and Thatte et al. (2021). However, AI surrogate models do not exhibit a significant speed advantage over traditional 1D performance prediction methods, and their physical interpretations are less explicit. The authors argue that the framework of traditional 1D performance analysis should still be retained. Instead, the powerful data processing capabilities of AI can be leveraged to transfer or expand some empirical loss models in traditional methods.

Model transfer is often used in situations where changes in certain design variables go beyond the applicability of traditional models. The authors (Liu et al. 2024) have presented an example of loss model transfer by AI. It is applied to 1D analysis of axial helium turbines. Since most available loss models are established for gas turbines and there is little knowledge about helium turbines, we have proposed a method to obtain loss models for helium turbines by transfer learning. As shown in Fig. 7a, knowledge transfer includes feature transfer, model transfer, and data transfer (pre-train). A new set of loss models is trained based on the CFD results of 102 helium turbines. Figure 7b compares the performance of the original model for gas turbines and the new model for helium turbines on the

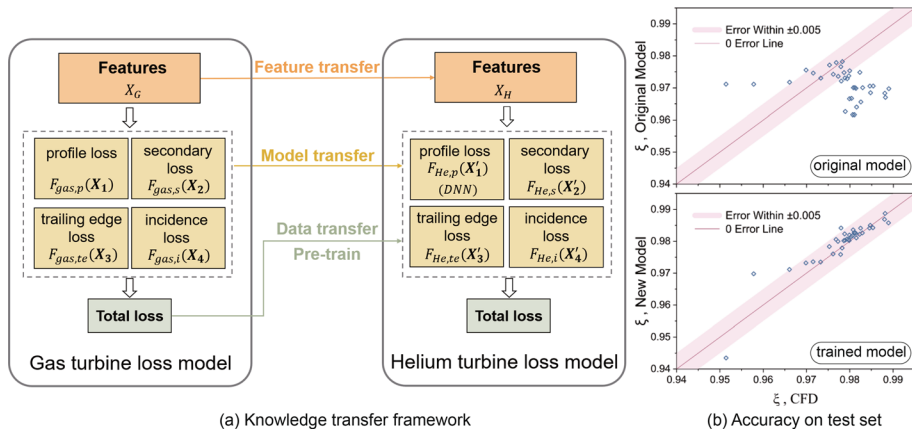


Fig. 7 Transfer learning of loss models for axial helium turbine: **a** knowledge transfer framework and **b** performance comparison of the original and the new model (Liu et al. 2024)

test set. The new model exhibits higher accuracy with prediction error within 0.005 for more than 90% of data points.

Model extension, on the other hand, is employed to analyze newly introduced design features or factors that were not considered in the original model. Yue et al. (2022) have presented an example of loss model extension by AI. Most classical loss models (Koch and Smith 1976; Wright and Miller 1992; Xu et al. 2021a) aim for the loss coefficient at the mid-span section, without taking the influence of secondary flow into account, thus losing the ability or accuracy to predict the spanwise distribution of the loss coefficient. Yue et al. provided an effective spanwise loss model with R^2 (coefficient of determination) of more than 0.8 on the test set. The modeling strategy is displayed in Fig. 8a. It introduced the secondary flow intensity (SFI) as an indicator to distinguish samples with different levels of secondary flows. A selector based on SVM was trained to classify the SFI levels. At each level, a dedicated loss estimator based on KNN was trained to predict the loss. To make the model more reliable, careful data cleaning and processing were carried out under the guidance of turbomachinery knowledge. As shown in Fig. 8b, the critical Mach number (Cumpsty 2004) was used as an indicator to eliminate samples with choked flows which all engineers try to avoid. The model was used to predict the spanwise loss coefficient distribution of E3's ten stators (Hollowny et al. 1982). General good correspondence is observed in the results (Fig. 8c), which is beyond the capability of classical loss models.

Similarly, some traditional theoretical models for turbomachinery aerodynamics such as the similitude theory (Xu and Zou 2022) and the boundary layer loss model (Kacker and Okapuu 1982) can be further expanded with the aid of AI to evolve into more universal models such as similitude methods considering real-gas effects (Xu et al. 2021b) and boundary layer loss models considering inflow unsteady effects (Simoni et al. 2020).

Q3D analysis

The Quasi-3D analysis is used to derive blade loading distributions, 2D flow fields, and aerodynamic performance based on given blade profiles and boundary conditions. Classical S1 and S2 analyses are performed by solving the differential Euler equations and radial equilibrium equations (Wu 1952), whereas AI can directly map the blade profiles to aerodynamic performance. It should be noted that most researchers (Pierret and Van

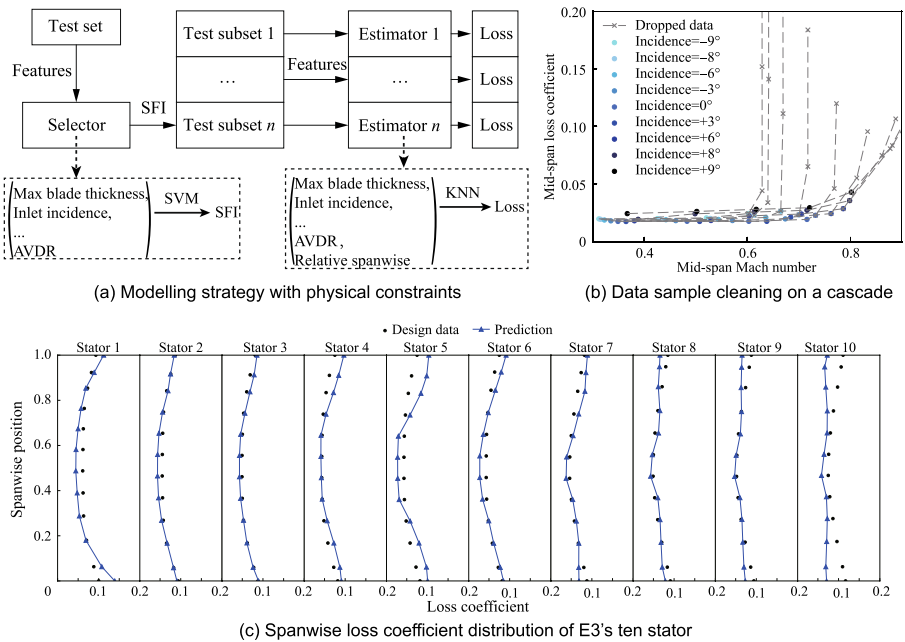


Fig. 8 AI extends classical loss models to predicting loss spanwise distribution: **a** data sample cleaning and processing, **b** modeling strategy, and **c** application to the E3's ten stators (Yue et al. 2022)

den Braembussche 1999; Du et al. 2022; Keane and Voutchkov 2022; Li et al. 2022b) do not recommend directly mapping geometric parameters to aerodynamic performance, as pure mathematical models lack physical universality. A common approach is to introduce intermediate physical quantities as constraints to the model. Taking axial turbines as an example, a two-step prediction model is provided by Du et al. (2022). As shown in Fig. 9b, the Dual Convolutional Neural Network (DCNN) consists of a field reconstruction network and a performance prediction network. In the analysis process, the distribution of temperature and pressure is first predicted by the field reconstruction network. Then the aerodynamic performance is predicted by the performance prediction network according to the output in the first step. The accuracy of the data-driven surrogate model is well improved by the appropriate incorporation of physical information. Feng et al. (2023) have also used velocity distributions as crucial physical information and recommend using loss coefficient instead of aerodynamic efficiency as the label.

In the aforementioned study, physical information is primarily embedded into the data-driven models in two ways: (1) Observational biases. A large amount of physical data containing key physical information, such as pressure/velocity distribution, is introduced during the training process. These observational data can reflect the underlying physical principles that dictate their generation, and, in principle, can serve as a weak mechanism to incorporate these principles into the data-driven model. (2) Inductive biases. The other approach is to design specialized neural network architectures that implicitly embed any prior knowledge and inductive biases associated with a specific predictive task. An example is the DCNN shown in Fig. 9b, which uses the loading distribution as the intermediate parameter between two tailored neural networks.

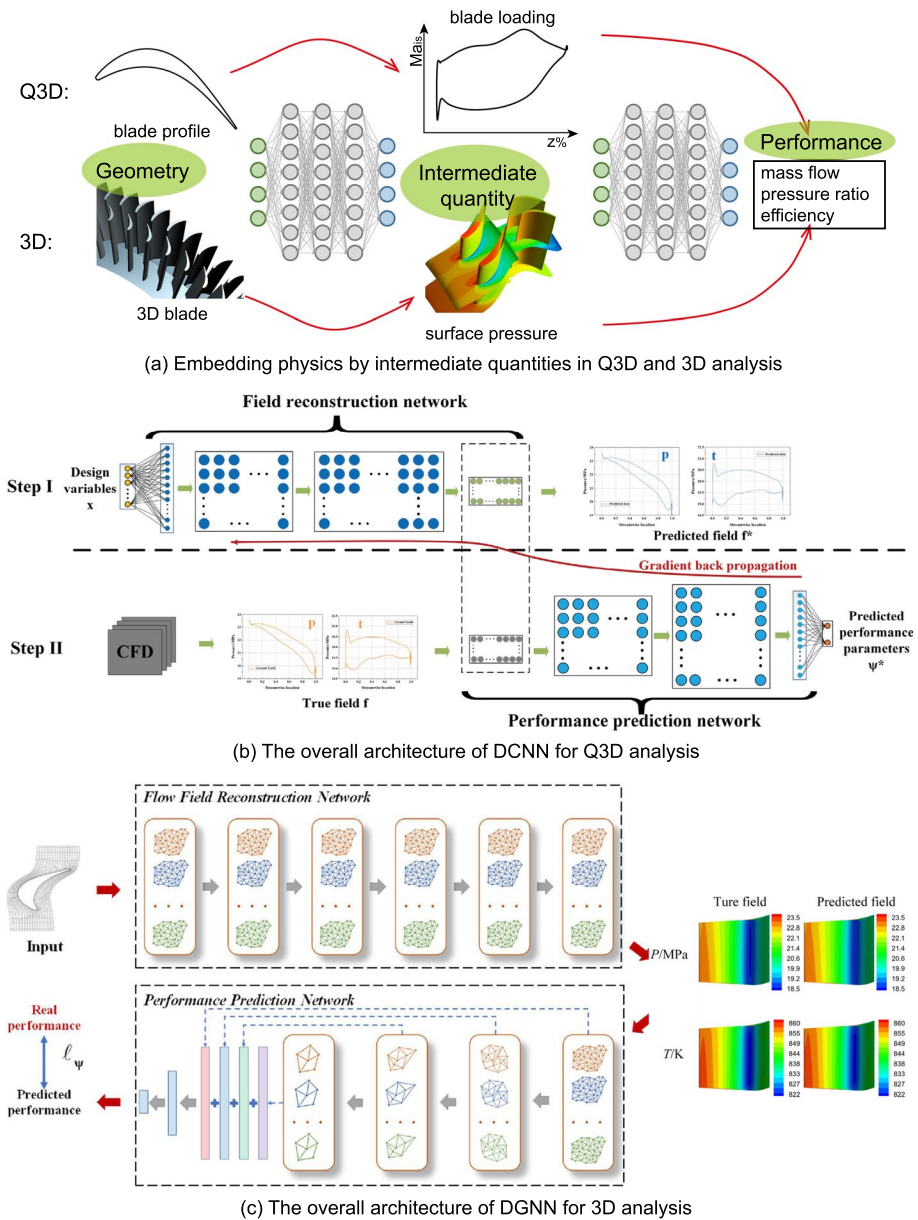


Fig. 9 **a** The idea of step-by-step mapping to embed physics by intermediate quantities. **b** The overall architecture of DCNN for Q3D analysis (Du et al. 2022). **c** Overall architecture of the DGNN for 3D analysis. A trained framework can predict the target flow fields along with the performances of a given turbine within 0.05 s. It could be a real-time simulation and analysis approach for turbomachinery (Li et al. 2022b)

3D analysis

Current design systems use high-resolution numerical simulations for 3D analysis, but they are computationally expensive when applied to optimizations. Data-driven surrogate

models can play the same role in a more practical and viable way. However, in such surrogate models, both the input parameters (3D blade geometry) and output parameters (3D flow fields) require data representation in extremely high dimensions. Using them directly for machine learning would result in an unacceptable increase in input and output dimensions. Therefore, it is necessary to first reduce the dimensionality of the data.

For the dimensionality reduction of 3D blade geometry, classical blade parameterization methods (Pritchard 2015; Kidikian et al. 2021) can be adopted. This approach represents the 3D blade by the design parameters of the multi-sections spanwise and the stacking curve. While this method is simple and well-established, it struggles with high feature dimensions in complex 3D geometries because of a dozen 2D blade sections. Another option is to employ machine learning techniques introduced in Sect. 2.3. These techniques are exceptionally versatile and capable of effectively handling arbitrary geometries.

Dimensionality reduction of 3D flow field data is a seldom-explored area, primarily owing to the vast amount of data inherent within fluid domains. Given that most 3D analyses do not meticulously scrutinize the entirety of flow field information, researchers often choose to retain only the relevant Physical Quantities of Interest (QoI) while discarding all other data. The authors have summarized the input features, QoI, and machine learning models in several studies, as listed in Table 3.

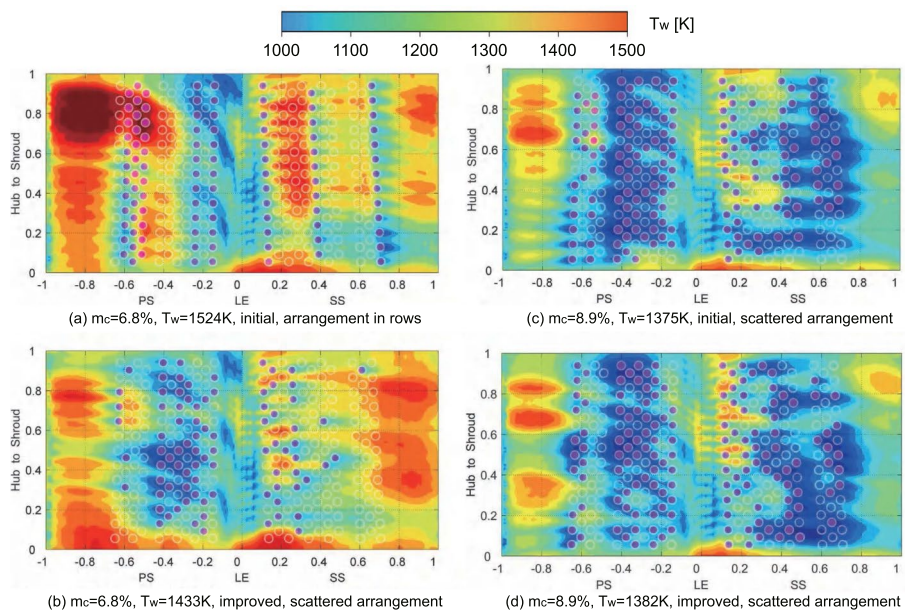
In the above studies, aerodynamic performance parameters are the most critical QoI that are the unanimous choice for all researchers. Some researchers directly establish a mapping between 3D blade geometry and aerodynamic performance parameters. However, as mentioned in the previous subsection about 2D analysis, this approach is not recommended. An intermediate parameter containing key physical information is strongly recommended, such as the blade surface distribution of pressure employed by Wang et al. (2021), Pongetti et al. (2021), and Du et al. (2022). Machine learning tools available for predicting 3D pressure distributions primarily rely on Convolutional Neural Networks (CNN) and Conditional Generative Adversarial Networks (cGAN). Both of them treat flow field data as equivalent to image data, establishing an end-to-end mapping between blade images and flow field images. However, the flow field inside a turbomachine is not a typical image, and pixelization inevitably leads to the loss of flow field information, especially in the near-wall flow region which is the focus of aerodynamic designers. This limitation can easily result in non-physical predictions. In this regard, Kashefi et al. (2021) developed a point-cloud deep learning framework which enables the end-to-end mapping between the spatial positions of each node in CFD meshes and corresponding physical parameters. As a result, the flow information in critical areas is well preserved. Another solution from Li et al. (2022b) is to leverage the inherent advantages of Graph Neural Networks in handling graph-structured data for predicting internal flow fields in turbomachines. As shown in Fig. 9c, they propose a Dual-Graph Neural Network (DGNN) that effectively utilizes data from all grid nodes and characterizes the neighborhood relationships of grids through the adjacency matrix. This approach efficiently captures near-wall flow information and generates relatively accurate flow field predictions for different operating conditions and 3D geometries.

There are also some noteworthy specialized techniques in the above studies. During the process of feature extraction, Pongetti et al. (2021) and Beqiraj et al. (2022) utilized dimensionality reduction through Autoencoders, considerably enhancing the freedom of the 3D blade design. During the process of modeling, Pongetti et al. (2021) and Du et al. (2022) employed the idea of step-by-step mapping. As shown in Fig. 9a, the 3D blade geometry is first mapped to its surface pressure distribution, and then both the blade geometry and its surface pressure distribution are used to predict the overall performance. As has been

Table 3 A brief survey of recent literature on AI-based 3D aerodynamic analysis for turbomachinery

Reference	Study object	Surrogate models	Features	QoI
Wang et al. (2021)	Axial turbine	cGAN	Parameterization of blade profile at hub, mid, shroud	Blade surface pressure distribution
Hu et al. (2022)	Axial compressor	SVM	Control points of Free-form Deformation (Sedberg and Parry 1986)	Isentropic efficiency, total pressure ratio, mass flow rate
Kosowski et al. (2009)	Axial steam turbine	FNN	Lean angle and sweep angle of twisted blade	Spatial distribution of entropy, pressure, velocity, and energy loss
Pongetti et al. (2021)	Axial compressor	AE, FNN AE, GP	Latent parameters of AE Blade surface pressure distribution, Latent parameters of AE	Blade surface pressure distribution Isentropic efficiency
Cheng et al. (2021)	Centrifugal impeller	SVM	Bezier control points of meridional flow path and blade	Isentropic efficiency at design point, pressure ratio at near stall point, and choked mass flow
Lopez et al. (2021)	Axial fan	FNN	Design parameters for sweep, lean, skew, edge recambering	Fan bypass isentropic efficiency
Du et al. (2022)	Axial turbine	DCNN	Bezier control points of blade profile	Blade surface pressure distribution, isentropic efficiency
Beqiraj et al. (2022)	Axial compressor	AE, Point net (Charles et al. 2017)	Structured grid-based point cloud of the 3D geometry	Isentropic efficiency, pressure ratio, temperature ratio, mass flow rate

Similar ideas can be employed for flow analysis inside the local 3D geometric structures, such as the casing treatment in compressors (Li et al. 2022c; Chi et al. 2023), the squealer tip gap (Maral et al. 2019) and the film cooling (Wang et al. 2022b; He et al. 2022a) in turbines. Taking the turbine film cooling as an example, He et al. (2022a) proposed a method for reconstructing the distribution of cooling efficiency on turbine blade surfaces using cGAN. They generated a database consisting of a wide range of flow field patterns with varying distributions of cooling holes. These flow field patterns were introduced as conditions into the discriminator network, and the difference between real and predicted values was captured through the adversarial loss derived from the competition between the Generator and Discriminator neural networks. The trained model was used to optimize the placement of cooling holes in a high-pressure turbine subject to inflow swirl and inlet hot streaks. As illustrated in Fig. 10, compared to uniformly distributed cooling holes, the optimized discretely distributed cooling holes reduced the maximum turbine blade surface temperature by approximately 100K under the same cooling air mass flow rate, significantly improving the cooling efficiency.



The optimization algorithm is a class of mathematical tools dedicated to obtaining optimal policy. They can be broadly divided into two categories: the first one is gradient-based optimization algorithms (Jameson and Martinelli 2000; Ruder 2017) rooted in rigorous mathematical theory, such as the steepest-descent method, Newton method, conjugate gradient method, Lagrange multiplier method, etc. These methods are applicable to quadratic programming and convex programming problems. They can guarantee global convergence if the objective function satisfies certain conditions. The other category is intelligent optimization algorithms based on the evolution of biological survival or natural physical phenomena, such as genetic algorithms (GA) (Goldberg 1994), differential evolution algorithms (Storn and Price 1997), particle swarm algorithms (PSO) (Kennedy and Eberhart 1995), ant colony algorithms (Alberto et al. 1992), simulated annealing algorithms (Steinbrunn et al. 1997), etc. These methods are simple to implement, and do not place any special demands on the analytic properties of the objective function. However, their convergence speed and stability depend on the specific algorithmic strategy. The design optimization of turbomachinery is usually a high-dimensional, nonconvex, nonlinear programming problem. Its objective function is a flow solver or the aforementioned AI-based surrogate models. There is no explicit expression for the objective function, making intelligent optimization algorithms more suitable for turbomachinery optimization. Currently, numerous intelligent optimization algorithms have been successfully applied to 1D (Elshamy et al. 2009; Bourabia et al. 2020; Bicchi et al. 2022; Dipierro et al. 2022; Li et al. 2024; Shahrabi Farahani et al. 2024), 2D (Sanger 1983; Li et al. 1997; Köller et al. 1999; Benini and Toffolo 2002; Shahrabi Farahani et al. 2023), and 3D (Lee and Kim 2000; Benini 2004; Oyama et al. 2004; Derakhshan et al. 2010; Kim et al. 2010, 2011; Okui et al. 2013) blade optimizations. Li et al. (2022a) provided an excellent review of machine learning in aerodynamic shape optimization. A typical optimization framework based on intelligent optimization algorithms is illustrated in Fig. 11a. In this optimization framework, AI-based surrogate models are employed to substitute the role of CFD, with the aim of accelerating the optimization process.

AI also offers unique solutions to optimization problems, specifically through deep reinforcement learning (DRL). DRL excels at handling nonlinear high-dimensional optimization problems, and its transferability can significantly reduce optimization time through

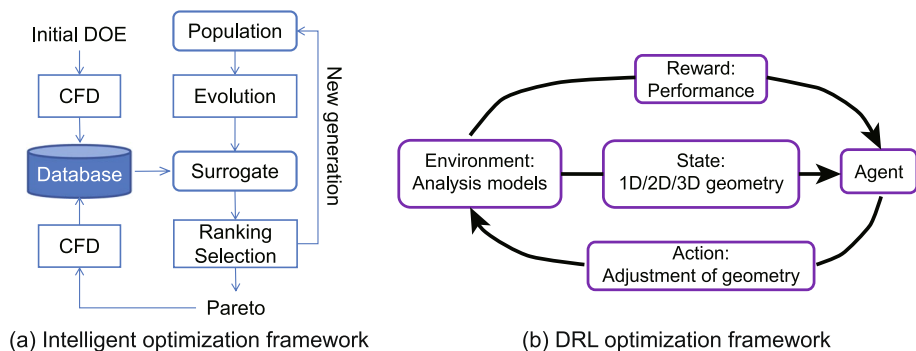


Fig. 11 Optimization frameworks for turbomachinery design. **a** Framework based on intelligent optimization algorithms. AI-based surrogate models are necessary to speed up the process. **b** Framework based on deep reinforcement learning. The performance optimization problem has been transformed into an AI autonomous decision-making problem with performance gain as the reward

transfer learning. When applied to turbomachinery, as shown in Fig. 11b, it is suggested to set the aforementioned surrogate models as the environment, the current blade geometry as the state, the adjustments to design parameters as the action, and the performance increment as the reward. Liu et al. (2022) employed the deep deterministic policy gradient (DDPG) algorithm (Lillicrap et al. 2019) along with a meanline analysis code in the optimization of a 9-stage high-pressure axial compressor with a pressure ratio of 5. From the design results, DRL did not exhibit any advantages over traditional optimization algorithms like GA. However, when Liu et al. utilized this model to optimize another compressor with a pressure ratio of 5.25, the DDPG method achieved an ideal result in just 8 steps, whereas GA required more than 200 iterations. Therefore, it is recommended to use DRL in 1D design optimization. Beqiraj et al. (2022) also recommended the Soft Actor-Critic method (Haarnoja et al. 2019) for 3D blade optimization.

3.3 Remarks

In the design mode, the key to the application of AI lies in the construction of databases. Turbomachines designed by AI mirror the design level of samples in the database. Only by timely upgrades of the database can we continually enhance the design capabilities of AI.

In the optimization mode, the key to the application of AI lies in the accuracy of analysis models. Only when the models conform to physical laws can the results be trusted. Since the computational speed advantage of AI surrogate models is more apparent in the 3D analysis, it is recommended to continue to use classical physics-based methods in the 1D and Q3D analysis, and only use AI to improve or extend empirical loss models. During the application of AI, it is essential to incorporate physical constraints appropriately by careful data cleaning and step-by-step mapping.

4 AI-improved validation method

Although assessments have been conducted in the previous section by AI surrogate models, it is believed that data-driven methods are always less reliable than physics-driven methods. Therefore, further validation through rigorous physical methods becomes imperative. Such methods include numerical simulations and prototype experiments. However, although physical laws play a dominant role in these methods, there are still some non-physical, empirical parameters that limit their practical engineering application. Specifically, in numerical simulations, direct numerical simulation (DNS) (Moin and Mahesh 1998) demands an immense amount of computing resources, making it unacceptable in engineering applications. As a result, Navier–Stokes equations are always reduced down to Reynolds-averaged Navier–Stokes equations (RANS), while an empirical turbulence model has to be introduced. In prototype experiments, due to limitations in measurement capabilities, designers must rely on their experience to infer quantities of interest that are challenging to measure directly. As depicted in Fig. 12a and d, these processes that rely on manual experience provide an opportunity for AI to intervene and help improve accuracy and expand functionality.

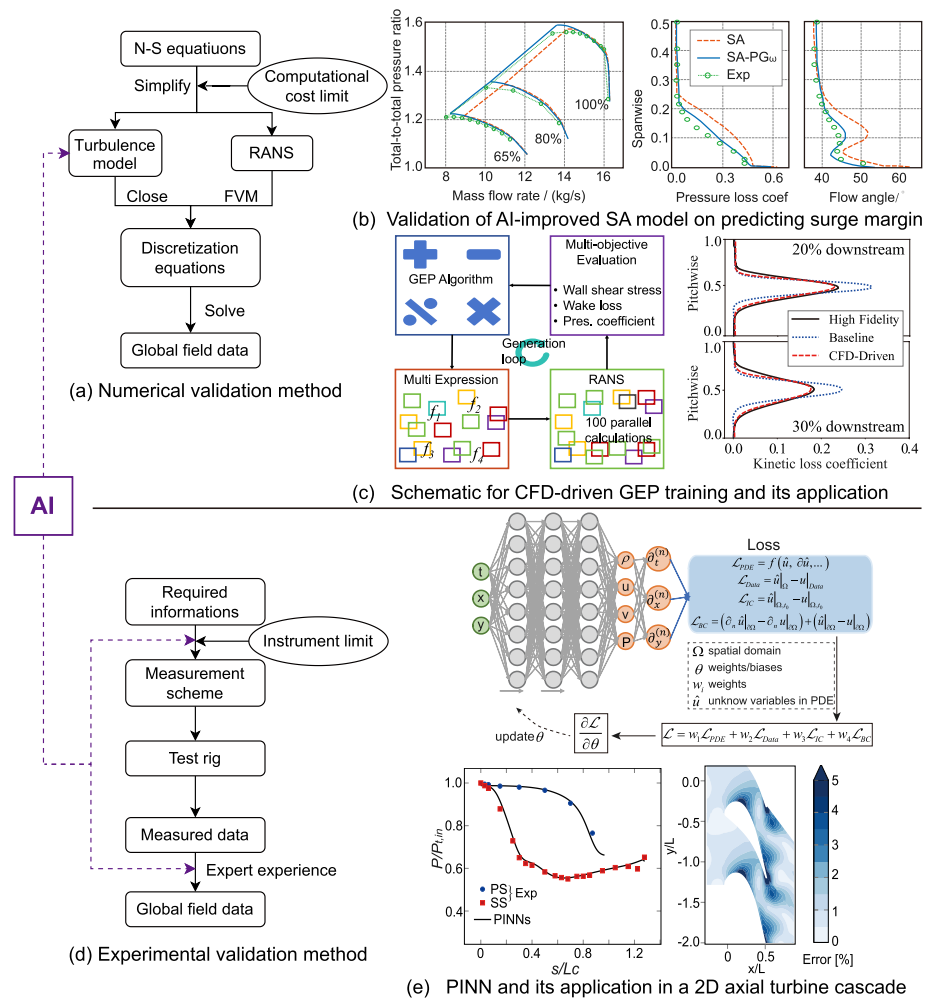


Fig. 12 **a** Process for numerical validation method, with the dash lines show procedures in which AI can get involved. **b** Prediction of surge margin and exit spanwise profile by a AI-improved SA model (He et al. 2022b). **c** Schematic for CFD-driven GEP training and its application to T106A turbine wake loss profile prediction (Zhao et al. 2020b). **d** Process for experimental validation method, with the dash lines show procedures in which AI can get involved. **e** PINN and its application in a 2D axial turbine cascade. PINN uses a fully-connected neural network, with time and space coordinates as inputs, is used to approximate the solution for the PDE. It is used to infer flow field from the measured blade pressure distribution of the 2D axial turbine cascade (Post et al. 2022)

4.1 AI application in numerical simulation: data-driven closure for RANS

Numerical simulation stands as an essential analysis tool for capturing full-field flow information in turbomachines (Wang et al. 2013). Due to the limited computing resources, RANS is still the most popular tool in engineering applications. The pivotal factor for achieving high accuracy within RANS lies in the choice of turbulence model (Durbin 2018). In the realm of turbomachinery aerodynamics, eddy-viscosity models (EVM) have

been widely used, including SA (Spalart and Allmaras 1992), $k - \epsilon$ (Jones and Launder 1972; Launder and Spalding 1974), $k - \omega$ (Wilcox 2006, 2008), SST (Menter 1994), etc. However, all these models follow the Boussinesq approximation which oversimplifies turbulence as homogeneous and assumes a linear relationship between Reynolds stress and strain rate. These oversimplifications yield inadequate predictions for complex flow structures in turbomachines, such as tip leakage, endwall vortices, boundary layer transitions, separation, shockwave-boundary layer interactions, etc (Hirsch and Tartinville 2009).

Nowadays, more advanced turbulence models can be learned from the data (Duraismay et al. 2019). In the modeling process, a high-fidelity database should be first established via high-accuracy experiments or high-resolution simulations (DNS, LES) (McConkey et al. 2021). This database must cover sufficient elementary flows such as separated flows, attached boundary layers, free shear flows, etc. Then features and labels should be determined either based on theoretical derivation or by referring to existing models (Yin et al. 2020). Finally, models can be established by machine learning.

There are two approaches to establishing a turbulence model. The first one acknowledges the theoretical framework of classical eddy-viscosity turbulence models and only refines some details of the control equations by adding source terms (Ferrero et al. 2020), correcting coefficients (Singh et al. 2017; Amstad et al. 2022), or constructing a deviation function for the Reynolds stress (Parish and Duraismay 2016; Singh and Duraismay 2016; Zhang et al. 2019). In these refinement models, the features are usually chosen from the characteristic parameters of original turbulence models, and the label is the selected correction item. An exemplary instance is the enhancement of the one-equation model $SA - HPG$ (Lee et al. 2018) by He et al. (2022b). This modified model ($SA - PG_{\omega}$) is specifically trained for axial flow compressors. The modification is based on the dimensionless vortical pressure gradient, which identifies blockage cells featured by 3D swirling, adverse pressure gradient, and low-momentum flows. It unblocks the compressor passage by enhancing the eddy viscosity in the identified blockage cells. The model coefficients are calibrated via Bayesian inference using the NASA Rotor 67 radial profile data (Strazisar et al. 1989) at peak-efficiency and near-stall points. As shown in Fig. 12b, $SA - PG_{\omega}$ exhibits improved predictions of stall margins and effectively captures the corner separation features.

The second one argues that the Boussinesq approximation leads to inherent structural errors, so the framework of classical eddy-viscosity models should be discarded. Although AI is able to directly establish mappings from turbulence characteristic quantities to Reynolds stresses (Fang et al. 2022), it may not guarantee the preservation of fundamental turbulence features such as Galilean invariance and scale invariance during the modeling process. Hence, some researchers (Ling et al. 2016; Xie et al. 2019; Zhao et al. 2020b) opt for an alternative framework: the explicit algebraic Reynolds stress model. This framework aims to establish mappings from nondimensional scalar invariants (Pope 1975) to Reynolds stresses. An illustrative example is the CFD-driven gene-expression programming (GEP) algorithm proposed by Weatheritt and Sandberg (2016), Zhao et al. (2020b), Waschkowski et al. (2022). This algorithm uses GEP to derive explicit expressions that are highly interpretable and applicable. Zhao et al. (2020b) utilized this algorithm to train a Reynolds stress model for the wake mixing of an axial turbine T106A (Stadtmüller and Fottner 2001). Results of the new model are shown in Fig. 12c. Significant improvements can be found in the prediction accuracy when compared to the classical baseline model.

In addition, some researchers have also explored the application of machine learning in the uncertainty assessment of turbulence models (Xiao et al. 2016; He et al. 2020) and the computation acceleration of numerical simulations (Kochkov et al. 2021).

4.2 AI application in experimental test: field inversion from test data

The experimental test is the most reliable tool for acquiring internal flow information in turbomachines. To date, CFD has not been able to completely replace the experimental tests. A performance test of any new components or whole machines in a suitable experimental rig is still the final arbiter of the aerodynamic design (Casey and Robinson 2021). Nowadays, frequently used measurement techniques for turbomachinery aerodynamics include contact methods like hot wires, pressure tubes, and thermocouples, as well as non-contact methods such as Laser Doppler Velocimetry (Osborne et al. 1974; Eckardt 1976), Laser-2-Focus Velocimeter (Krain and Hoffman 1989; Ziegler et al. 2003), Particle Image Velocimetry (PIV) (Wernet et al. 2000; Westerweel et al. 2013), etc. However, all these techniques can only obtain information about a single physical quantity at a single point or on a single plane. To address this limitation, a series of enhanced techniques have been developed based on PIV, such as dual-plane PIV (Ganapathisubramani et al. 2005), holographic PIV (Hinsch 2002), and tomographic PIV (Elsinga et al. 2006). Nevertheless, due to the narrow blade passages in turbomachines, these methods can only measure the velocity of sparse 3D points. The challenge of achieving high-resolution measurements of the whole 3D flow field remains unresolved.

This issue can be resolved by physics-informed neural networks (PINNs) (Raissi et al. 2017a, b). In essence, PINNs is a solver that uses neural networks to approximate the solution of the partial differential equation(PDE). As shown in Fig. 12e, by incorporating the residual of control equations into the loss function, the equation-solving problem is transformed into an optimization problem to minimize the loss. Since measured data can be naturally inserted into the solver as losses, PINNs outperform classical solvers when dealing with inverse problems (Raissi 2018; Karniadakis et al. 2021). Therefore, PINNs are well adapted for flow field inversion in experiments. A typical example has been proposed by Wang et al. (2022a). Dense velocity and pressure fields of the 3D wake flow of a hemisphere have been reconstructed from sparse data measured by tomographic PIV (Elsinga et al. 2006). The horseshoe vortex, which is missing in the raw PIV data, can be distinctly displayed after being processed by the PINNs. Post et al. (2022) has presented an application of PINNs in turbomachinery. As shown in 12e, full-field flows inside a 2D axial turbine cascade can be inferred from the measured pressure distribution with errors no more than 5%.

4.3 Remarks

Physics-driven methods should be the final arbiter of a new design. AI can serve as a supplement to fill the gaps of classical physics-driven methods. For numerical simulations, AI can refine turbulence models for RANS and improve the solution accuracy. For experimental tests, AI helps invert global flow fields from sparse measuring data. The extended functions of classical measuring techniques may reshape the fundamental principles of some aerodynamic experiment schemes.

5 AI-assisted maintenance system

Maintenance is the process of preserving the good condition of a system for the purpose of reducing operating and maintenance costs while at the same time assuring maximum operating time and achieving the highest possible production rate. For turbomachinery, there are three maintenance philosophies (de Castro-Cros et al. 2021): reactive, preventive, and predictive maintenance. Reactive maintenance is performed only when turbomachines break down (Wang et al. 2014). Since the failure may have already caused irreversible damage, this run-to-failure repair is less effective and unplanned machine downtime is very costly. Preventive maintenance (Alsyof 2007) is performed according to a determined schedule planned in advance, regardless of the health status of turbomachines. It is widely practiced today, but repetitive maintenance routines may lead to potential waste of half-life spare parts and excessive disassembly may create additional risks to a healthy system (Forsthofer 2011). The most effective way to perform any maintenance activity is to thoroughly plan that activity based on condition changes to equipment. This is referred to as predictive maintenance (Pusey 2007). It is a proactive process that can predict trends, behavior patterns, and correlations by statistical or machine-learned models, with the aim of estimating pending failures in advance to improve the decision-making process for the maintenance activity avoiding mainly the downtime (Carvalho et al. 2019; Zonta et al. 2020).

Artificial intelligence has emerged as a potent tool for developing intelligent predictive algorithms for the predictive maintenance of turbomachines. During the operation of turbomachines, AI monitors health-relevant parameters such as flow data, vibration data, noise data, oil data, and other nondestructive inspection data. These data are used for health status identification and anomaly detection (Stetco et al. 2019; Yusoff et al. 2019). Based on trained models, AI can perform online real-time prognoses (predicting a future condition on the basis of present signs and symptoms) (Hipple et al. 2020) and offline diagnoses (identifying failure mode from its signs or symptoms) (Lv et al. 2021; Menga et al. 2022). Based on the results, AI can also provide decision support for the maintenance. The data-driven models are the core of AI's application in turbomachinery maintenance. This section will introduce how AI is applied in three typical maintenance scenarios.

5.1 Compressor stall warning

Aerodynamic instability has been a problem as long as turbomachinery compressors have been built. Compressor instability includes the rotating stall and the surge. The rotating stall is a local unsteady stalled condition rotating within the blading. The surge is a much larger global disturbance involving the whole compression system. It is destructive, placing a heavy mechanical burden on the compressor structure and leading to serious power losses. Since the occurrence of the surge is often accompanied by the rotating stall (Greitzer 1981), it is possible to predict aerodynamic instability by detecting and recognizing the stall inception.

Stall inception is typically detected according to the fluctuating characteristic signals (usually collected by arrays of circumferentially spaced hot wires or high-frequency response pressure transducers) and recognized by signal analysis. Stall inception data could be analyzed in the time domain (Park 1994; Cameron and Morris 2007), frequency domain (Longley 1988; Tryfonidis et al. 1995; Hönen and Arnold 2003), or both of them (Dremin et al. 2002; Li et al. 2011). In the time domain, inception is recognized by the amplitude, variance, or spatial correlation of characteristic signals. These methods are simple but

unreliable due to the noisy data. In the frequency domain, inception is recognized by power spectral density, traveling wave energy, or spectrum of the harmonic modal wave. These methods are relatively more accurate but the signals must be stable and complete. More recently, wavelet methods and Hilbert-Huang transform have been applied as more reliable techniques. Nevertheless, these methods still demand complete signals before and after the stall occurs. Therefore, they are only suitable for offline analysis after the stall happens, but cannot achieve online real-time stall warnings.

The stall warning is essentially a feature-identification problem and deep learning has been widely applied to this kind of problem successfully. The recognition model for stall inception could be trained by the following procedures: (1) collect characteristic signals in the whole stall process and make pre-processing such as filtering and resampling; (2) slice the time series data by small time steps and label each segment according to the analysis results of traditional methods like wavelet decomposition; (3) train the DNN by labeled data.

In reviewing the literature, no unified standard was found for the selection of feature signals, data preprocessing methods, and machine learning models. Zhao et al. (2020a) chose the casing pressure at the rotor inlet as the characteristic signal and a dilated causal convolutional neural network as the model. The occurrence of a progressive stall was successfully predicted 327 ms in advance. Lin et al. (2019) chose to use the flow coefficient combined with a radial basis function network. They successfully achieved 8.8 rotor revolutions in advance to give warning of an abrupt stall in the low-speed axial compressor test at Beihang University, as shown in Fig. 13a. Other researchers also used long short-term memory networks (Hipple et al. 2020) or deep residual networks Zhang et al. (2022) to conduct stall warnings.

The above studies indicate that AI has the capability to timely and accurately identify the stall inception of axial compressors, holding significant promise for stall warnings. However, due to the lack of stall inception data, it is challenging to guarantee the generalization of current models. In addition, existing research has primarily focused on single-stage axial compressors, whereas real applications often involve multi-stage axial compressors and centrifugal compressors. Further study is required to explore how to apply AI for stall warnings in these scenarios.

5.2 Early defect detection

Since turbomachines usually work in harsh environments caused by high rotational speed, high pressure, and high temperature difference, they are easily prone to blade fatigue in long-time operations. A minor defect, if not detected in time, may develop into a fracture failure. The broken blade may hurt the rear blades and other components, resulting in catastrophic damage. Therefore, blade defect detection is a crucial aspect of turbomachinery maintenance. It falls within the framework of AI-assisted maintenance and consequently can be categorized as offline techniques and online techniques.

Offline defect detection involves inspecting blades when the turbomachine is halted. Classical non-destructive inspection techniques include eddy current testing (Sasi et al. 2004), ultrasonic testing (Ageeva et al. 2013), magnetic testing of particles (Zou 2020), and radiography (Wong et al. 2006). These diagnoses fundamentally rely on mathematical tools such as wavelet analysis for feature recognition and anomaly detection in the collected signals. The exception is visual inspection, which utilizes a borescope to inspect the inner part of turbomachines without disassembling the component. This technique is widely

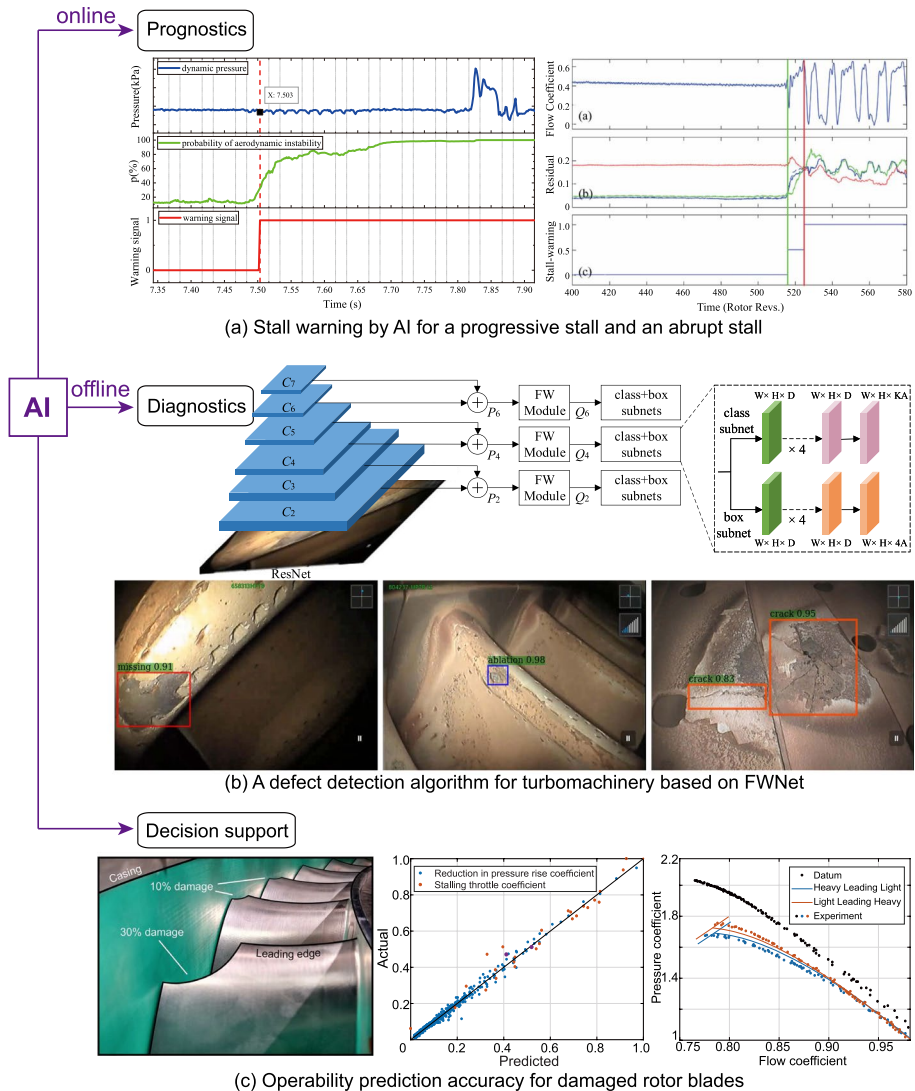


Fig. 13 **a** An example for prognoses. Compressor stall warning by AI for a progressive stall (Zhao et al. 2020a) and an abrupt stall (Lin et al. 2019). **b** An example for diagnoses. A fast and accurate defect detection algorithm for turbomachinery based on FWNNet (Chen et al. 2020). It can provide a diagnostic basis for professional inspectors. **c** An example for decision support. AI can accurately predict the operability of damaged rotor blades to determine whether maintenance is necessary (Taylor et al. 2020)

used because it avoids the reliability issues associated with repetitive disassembly. However, manual image inspection relying on experienced inspectors makes the entire process highly inefficient and limited to subjective individual judgments. With the advancement of computer vision and image processing, manual inspection is gradually being replaced by AI. Researchers (Kim and Lee 2019; Shen et al. 2019; Chen et al. 2020; Wong et al. 2021; Jaeger et al. 2022; Li et al. 2022d) have developed various neural networks with

CNN as the core for extracting blade defect features, such as Feature Pyramid Networks (Lin et al. 2017), Regional Convolutional Neural Networks (Bharati and Pramanik 2020), YOLO (You Only Look Once) (Jiang et al. 2022), etc. An exemplary work by Chen et al. (Chen et al. 2020) introduced a fast and accurate feature weighting network (FWNet) based on widely used CNNs and feature pyramids, as depicted in Fig. 13b. This model incorporates a feature weighting module for recalibrating channel-wise attention and enhancing the weights of valid features. It utilizes a CNN for hierarchical defect extraction, with a feature weighting module (FWM) recalibrating feature maps to improve feature propagation. A feature pyramid is constructed on top of the ResNet architecture, and task-specific subnetworks are employed to identify defects at various scales. The model was evaluated on a dataset comprising 1916 images with three defect categories, achieving a mean average precision of 89.4%.

Online defect detection, on the other hand, involves predicting blade defects in real-time operation based on the data of condition monitoring. Similar to the feature identification problem of stall warning in the previous subsection, AI is applied to online blade defect detection by training machine learning models with labeled feature signal datasets. The challenge is in the selection of characteristic signals (Huang et al. 2021). When damage appears in turbine blades, it causes changes in the blade vibration parameters. The most immediate change is the natural frequency (Rani et al. 2019). However, it is hard to measure the natural frequency of rotational blades separately during the operation, so this parameter is not widely used in practical applications. Since the vibration signals of blades can propagate to the rotating shaft, some researchers have used signals of angular velocity (Gubran and Sinha 2014) and shaft torque (Maynard and Trethewey 2000) as feature parameters for defect detection. In addition, a defect may also alter the aerodynamic state of the internal flow, so that defect detection can also be performed from this perspective (Libeyre et al. 2021).

The application of AI technology in offline blade defect detection is relatively mature, but in online blade defect diagnosis, it is still in the early research stage. More future research is demanded on how to select an appropriate characteristic signal, how to locate the defect on the blade, how to generalize these methods to different types of turbomachines, and so on.

5.3 Operability prediction

The aerodynamic performance of turbomachines is important to overall cycle performance. Although the standard aerodynamic performance has been measured in performance tests, it is likely to deteriorate during long-time operation when mechanical damage occurs such as corrosion, fouling, and abrasion. An accurate prediction of the performance degradation needs to be made quickly in order to sentence the hardware and minimize disruption.

The study of Taylor et al. (2020) confirms the possibility for AI to predict the operability of damaged compressors. With the help of rapid tests and physical parameters derived from engineering wisdom, two DNNs were trained based on 125 cases. They were able to predict the performance and surge margin of damaged compressors with an accuracy of 2% in a 95% confidence interval, far better than was possible by even the most experienced compressor designers. Figure 13c provides an illustration of the damaged blades and prediction accuracy. A similar method has been proposed by Friso et al. (2021) to predict the performance degradation of axial turbines in fouling conditions. AI can also be used to estimate the performance improvement specifically brought on by the compressor wash

(Raghavan et al. 2020). Trade-offs between the cost and benefit of the compressor wash can be carefully addressed with the aid of these predictions, and maintainers can make decisions on whether or not to perform compressor wash according to the result.

5.4 Remarks

This section only reviews the above three problems due to the fact that AI is currently only studied in these scenarios. It is evident that the “signal acquisition—feature recognition—fault prediction” workflow can also be applied to other maintenance problems of turbomachinery. Related research is yet to be conducted in the future. Furthermore, there is potential to establish a comprehensive intelligent turbomachinery maintenance system that can be applied to the prognoses and diagnoses of all fault issues during the life-cycle operations of turbomachines. Such a system could possess the capability of preemptive prediction and post-event diagnosis, and might even assist maintainers in generating maintenance plans.

6 Framework of AI application in turbomachinery aerodynamics

Based on the above discussion of AI applications in turbomachinery aerodynamics, Table 4 summarizes the research objectives, key scientific challenges in each phase, and the corresponding solutions brought by AI.

In light of the aforementioned characteristics of turbomachinery aerodynamics, in conjunction with the literature review in the latter sections and the author’s own experience, this paper synthesizes a framework in Fig. 14 to illustrate how AI can participate in aerodynamic tasks throughout the whole life cycle of turbomachines. Similar to aerodynamic researchers, AI is mainly involved in three areas: (1) aerodynamic design; (2) validation; (3) maintenance.

6.1 Application object

In the aerodynamic design part, AI takes the lead in the entire design process. The database should store adequate data about well-designed turbomachines, including geometry, design targets, and performance. AI automatically extracts features from the geometry and design targets, and then establishes a direct or indirect mapping from the features to the aerodynamic performance. This enables AI to acquire relevant knowledge for turbomachinery performance analysis, such as loss models and loading prediction models. In addition, for each set of design targets, the optimal aerodynamic performance and the corresponding geometry representing state-of-the-art design level need to be stored. AI establishes a mapping from the design targets to the best geometry, thereby acquiring the necessary knowledge for inverse design of turbomachinery, such as velocity-triangle design models and profile design models. AI-based aerodynamic design combines both design and analysis processes from low to high dimensions sequentially.

In the validation part, AI only acts as a supplement to classical methods such as numerical simulations and experimental tests. When applied to numerical simulations, AI extracts knowledge from high-fidelity data in DNS simulations and experimental measurements to improve turbulence models and transition models for RANS simulations. When applied to

Table 4 The scientific challenges in the development of turbomachinery and AI applications

R & D phases	R & D objectives	Scientific challenges	AI applications
Aerodynamic design	To find a blade geometry that can meet the design targets	High geometric design freedom makes the quality of the design results heavily reliant on the experience of the designers	Leveraging AI with a historical design database to map low-dimensional design targets to high-dimensional blade geometry
Aerodynamic optimization	To seek the blade geometry that maximizes aerodynamic performance under given constraints	The aerodynamic performance analysis relies on CFD, which comes with high computational costs and long optimization cycles	Training neural networks with carefully curated CFD data, using the concept of step-by-step mapping to embed physics, ultimately achieving rapid and accurate aerodynamic performance analysis in the form of a surrogate model
Numerical validation	To solve the N-S equations and obtain the aerodynamic performance	Trade-off between accuracy and speed. The commonly used RANS method in engineering offers speed but introduces empirical turbulence models	AI acquires high-precision turbulence models from high-accuracy experiments or high-resolution simulations (DNS, LES)
Experimental validation	To obtain the aerodynamic performance of turbomachines through measurements of key physical quantities	Limitations in measurement space and techniques make it difficult to obtain information of the whole flow field	Leveraging the advantages of PINN, it is possible to invert the whole flow field through sparse data, addressing inverse problems effectively
Maintenance	To predict impending defects and diagnose existing faults; To assess the current state of the turbomachine, forecast operability, and guide maintenance	Existing methods face challenges in extracting crucial information from high-dimensional condition monitoring data for prognosis and diagnosis; Assessing the state and predicting operability heavily relies on manual expertise	AI identifies fault characteristics from massive condition monitoring data, assesses the current turbine state, and predicts operability based on historical data

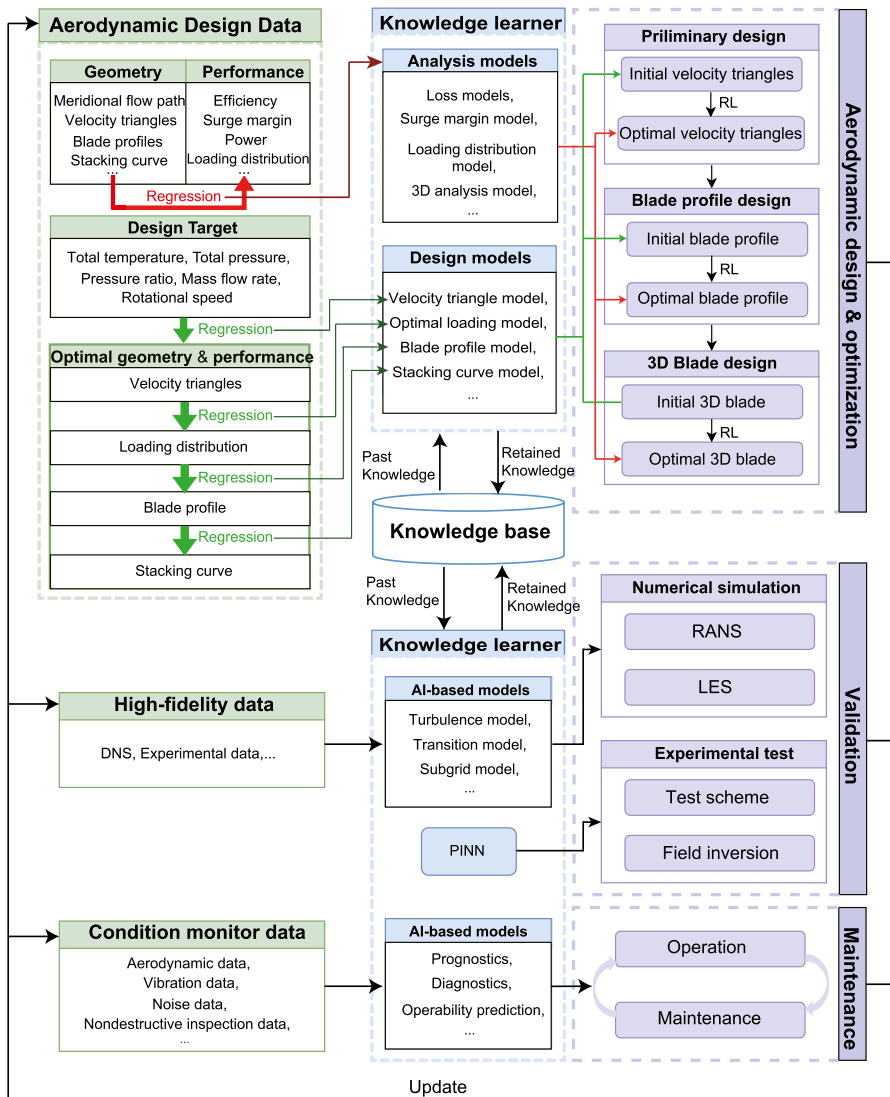


Fig. 14 A framework for the AI-based turbomachinery R & D system. The entire framework consists of three major components: **a** AI-based aerodynamic design system. This system, with AI at its core, leverages the advantages of AI in knowledge acquisition and autonomous decision-making to integrate them into the process of design and optimization. **b** AI-improved validation method. Validation relies on high-fidelity physics-driven methods which can be further enhanced by AI. **c** AI-assisted maintenance system. In this system, AI is equipped with functions for real-time monitoring of the operational status, fault prediction, and decision-making for maintenance plans

experimental tests, AI helps reconstruct overall flow fields from limited measurement data by physics-informed neural networks (PINNs).

In the maintenance part, AI serves as an assistant whose primary duties include real-time monitoring of turbomachines' operational status, providing early warnings for potential failures, and diagnosing and guiding maintenance procedures in case of abnormalities.

6.2 Application process

AI applications in the context of turbomachinery aerodynamics primarily follow a standard workflow of data-driven modeling:

The first step involves data collection and preprocessing. Typically, researchers should first gather a sufficient amount of data that encapsulates the desired knowledge. Subsequent data preprocessing encompasses data cleaning, feature selection, and standardization/normalization. Data cleaning serves the dual purpose of simplifying the dataset and achieving sample balance to prevent model performance imbalances caused by local sample scarcity or absence. Standardization and normalization aim to mitigate numerical scale differences among different input features, ensuring data generality. Feature selection relies on the researcher's own physical knowledge to identify independent variables relevant to the model output, enhancing model performance.

The second step is model selection. Available models include but are not limited to all the models listed in Sect. 2.3. Researchers need to choose the most suitable machine learning model based on the characteristics of the aerodynamic task. In some cases, tailored model architectures may be required to achieve optimal model performance. After determining the dataset and the machine learning model, researchers should divide the data into training and validation sets in a certain proportion, and then feed them into the machine learning model for training.

The final step is model post-processing. In practical engineering applications, it is often necessary to apply post-processing to the model's predictions to avoid unphysical solutions. For instance, specific physical constraints may be added to address particular problems, and threshold values may be set based on the boundaries of the training samples. Additionally, probabilistic analysis can be used to assess the uncertainty of model predictions, aiding researchers in better evaluating the credibility of the results.

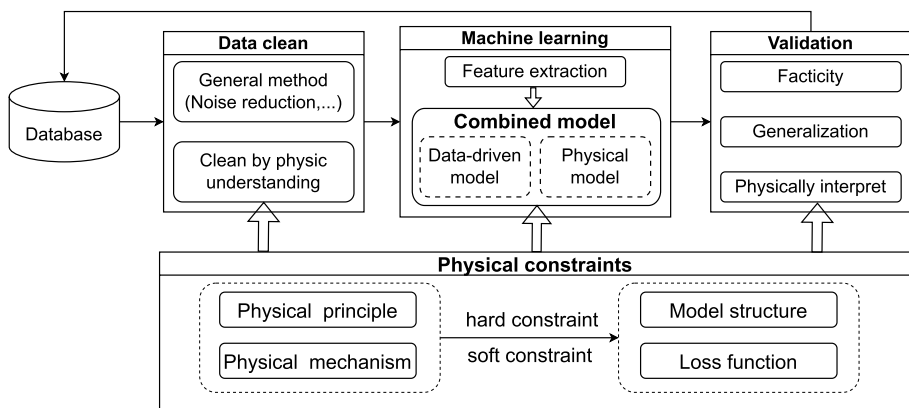


Fig. 15 Physical constraints should be incorporated during the application of AI for science

In addition to the general workflow outlined above, when applying AI to turbomachinery aerodynamics, special attention must be given to physical constraints. As illustrated in Fig. 15, physical information is embedded into the models through three pathways:

1. In the stage of data preprocessing, observational data ought to be thoroughly cleaned until it can reflect the underlying physical principles, and, in principle, can be used as a weak mechanism for embedding these principles into an AI model during its training phase. A typical process of the establishment and clean for a compressor blade profile database can refer to the literatures (Schnoes and Nicke 2017; Yue et al. 2022).
2. In the stage of knowledge learning, the selection of feature parameters should be based on aerodynamic knowledge relevant to turbomachinery. NN architectures should also be tailored to implicitly incorporate any prior knowledge. In certain cases, physical constraints can be flexibly imposed by appropriately penalizing the loss function of conventional NN approximations.
3. In the stage of validation, researchers should explore the underlying physics of the model. By interpreting the model, they can uncover the reasons for AI's success, analyze failures for potential improvements, and even potentially facilitate the discovery of new scientific laws.

6.3 Features of this framework

The proposed framework in Fig. 14 has the distinction of the following features:

1. Contrary to most studies that treat AI merely as a mathematical tool used to generate surrogate models for aerodynamic analysis and uphold the decision-making position for mankind, this paper argues that AI could dominate the whole aerodynamic design process. AI's prowess in knowledge learning and decision-making enables autonomously generating solutions without heavy reliance on human engineering experience.
2. The proposed framework incorporates the idea of lifelong learning which allows for self-upgrading. Four functional components have been established accordingly: (a) Database which keeps storing available data; (b) Knowledge learner which discovers knowledge with the help of past knowledge; (c) Knowledge base which stores previously learned knowledge; (d) Application which puts knowledge into practice.
3. This framework adopts the principles of AI4S. The modeling process must be guided by the physical laws of aerodynamics and thermodynamics in turbomachinery. This idea is manifested in the step-by-step aerodynamic design, which avoids directly mapping design objectives to 3D geometry. Instead, turbomachines are built from 1D to Q3D and then to 3D, with thorough optimization in each dimension to ensure that the results are physically correct. Meanwhile, design and analysis models in each step should also be constructed in the framework of AI for science as shown in Fig. 15.

7 Challenges

The reviewed studies provide promising progress in the application of AI technology in the field of turbomachinery aerodynamics. It must be understood, nevertheless, that such applications are still in the early stage of exploration. There is still a considerable gap in establishing a fully autonomous AI-based turbomachinery R & D system. Information

from papers published in this field and the authors' own insights suggest the following challenges:

I. Balance the data and physics in turbomachinery

- How to appropriately incorporate physical constraints in the modeling? Aerodynamic tasks for turbomachinery inherently adhere to the fundamental physical laws of aerodynamics and thermodynamics. Data-driven models built for them must also conform strictly to these laws, so physical constraints must be incorporated during the modeling process. In the research reviewed in this paper, physical constraints have been incorporated in different ways such as data cleaning, feature selection, step-by-step mapping, physics-informed loss functions, etc. However, there is a notable absence of dedicated research specifically addressing the issue of embedding physical constraints.
- How to understand the underlying physics behind data-driven models? Models obtained through data-driven methods predominantly capture the statistical characteristics of the training dataset and, on the surface, appear as mathematical models. However, since aerodynamic tasks of turbomachinery are fundamentally physical problems, models developed for these tasks inherently encompass deeper physical mechanisms. Gaining insights into the physical mechanisms behind the model can enhance researchers' understanding of turbomachinery aerodynamics and inspire novel ideas for new aerodynamic tasks.

II. Improve the AI-based aerodynamic design system

- How to appropriately apply AI to the inverse design of 3D blades? At the current stage, research has indicated that AI has the potential for rapid inverse design of 3D blades, but there is still room for improvement. How to consider 3D viscous effects when designing the stacking curve? How to prescribe a valid pressure distribution as the input for 3D inverse design? How to predict the possible flow choking and separation in the 3D analysis? How to account for the interaction between stages in multi-stage compressors and turbines? Addressing these issues requires further research and development in the application of AI to turbomachinery 3D design and optimization. It involves a multidisciplinary approach combining expertise in AI, fluid dynamics, and turbomachinery engineering.
- How to reduce the computational costs? For 3D analysis models, training databases should be constructed by CFD. For AI-improved simulations, high-fidelity databases should be established by LES or even DNS. Successful learning requires a large amount of high-quality training data, typically tens of thousands of samples. The computational cost is so high that it must be considered how to reduce the costs in large-scale engineering applications.
- How to assess the uncertainty of the results produced by AI? To some extent, data-driven models can be regarded as statistical models. The data distribution in the training set may not accurately represent the true distribution, leading to potential generalization issues in machine learning models. It is important for engineers to obtain confidence intervals for the predictions. How to quantify the uncertainty to help researchers determine their confidence in the results is one of the key concerns in the future.
- How can AI be used to guide aerodynamic experiments? PINNs exhibit the potential to reconstruct the entire flow field from limited measuring points, which greatly extends the capabilities of traditional measuring techniques. It has the potential to reshape the

fundamental principles of aerodynamic experiments in the future. However, up till now, there have been few studies that apply PINNs to the 3D viscous flows in turbomachinery and no mature solutions for aerodynamic experiments based on PINNs. Further research is needed in this direction to advance the field.

III. Develop the AI-assisted maintenance system

- How can AI be leveraged for real-time intelligent active control in turbomachines? In addition to aerodynamic optimizations, many other aerodynamic issues for turbomachinery also involve autonomous decision-making, such as active surge control of compressors, pulsation injection of turbine coolant, control laws of the whole cycle, etc. DRL holds great promise for addressing these types of active control challenges. Research in this area is poised for comprehensive exploration and development.
- How to build an intelligent turbomachinery maintenance system based on AI? The application of AI in turbomachinery maintenance is still in its early stages. There is currently a lack of a comprehensive turbomachinery maintenance system that can address all operational issues. Such a system should encompass four key aspects: condition monitoring, prognoses, diagnoses, and maintenance decision-making. It is actually a matter of multiple disciplines, including turbomachinery aerodynamics, structural strength, rotor dynamics, and reliability analysis. Meanwhile, the health status of turbomachines is closely related to other components of the entire cycle, so the intelligent maintenance system must be established from the perspective of the entire machine. This is a significant multidisciplinary research direction in multiple fields.

8 Conclusion

This paper provides a comprehensive overview of recent developments for the application of AI in turbomachinery aerodynamics, focusing on the role played by AI and the solutions developed by AI when applied to various aerodynamic tasks. Additionally, the authors strive to present a holistic view of the AI-based turbomachinery R & D system and identify some key challenges based on our own insights. Several conclusions can be drawn as follows:

- AI innovates a new research paradigm for turbomachinery aerodynamics. Its capabilities in data mining, feature extraction, knowledge acquisition and transfer enable it to learn from historical data, thus partially replacing the role of aerodynamic researchers in aspects such as aerodynamic design, validation, and maintenance. Simultaneously, AI provides data-driven solutions that offer a unique opportunity to generate understanding of previously intractable problems, which makes it a perfect complement to classical research methods. The AI-based turbomachinery R & D system represents a crucial avenue for future advancements in the field.
- In the aerodynamic design process, AI will play a leading role. Physical constraints must be well integrated into data-driven models to ensure the reliability of solutions. Several methods for incorporating physical constraints are available, which include: data cleaning using knowledge of turbomachinery, feature selection based on physical theories or engineering experience, step-by-step mapping with intermediate validation, and using physics-informed loss functions.

- In the aerodynamic validation process, ensuring the accuracy and reliability of performance validation demands a preference for physics-driven methods such as numerical simulations and experimental tests. AI should be employed primarily as a supplementary and improvement tool for addressing weaknesses within these physics-driven approaches. For numerical simulations, AI can refine turbulence models for RANS and improve the simulation accuracy. For experimental tests, AI helps invert global flow fields from sparse measuring data.
- In the maintenance process, AI serves as an assistant whose primary duties include monitoring the operating status of turbomachines, forecasting and alerting potential faults, diagnosing failures, and proposing maintenance schemes. The availability of extensive data from online monitoring and advanced inspections bridges the accuracy gap that existed in traditional preventive maintenance techniques and resolves previously intractable problems such as operability predictions from the engineering practice level.
- At present, the application of AI in turbomachinery aerodynamics is still at an early stage of exploration. There is a notable lack of both breadth and depth in research concerning various aerodynamic issues in turbomachinery. At a macro level, the development of a comprehensive AI-based turbomachinery R & D system is still evolving. At a micro level, many detailed investigations remain unexplored or require further in-depth research in areas such as 3D inverse design, experimental tests, active control, intelligent maintenance, etc. The field awaits the active participation of more researchers in these endeavors to advance our understanding and application of AI in turbomachinery aerodynamics.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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